



Simulation of droplet bouncing on flexible substrate in 2D and 3D with WC-TL SPH method

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ABSTRACT

Droplet impacts on super-hydrophobic surfaces play an important role in many fields. Previous studies have focused on rigid substrates and ignored the effect of substrate deformation. Simulating droplet impacts on a flexible substrate is a challenge to numerical methods due to the coupling of free surface, elasticity, large deformation and surface tension. In this study, a fully meshfree model of fluid–solid interaction is developed based on the smoothed particle hydrodynamics (SPH) method. The liquid droplet is modeled by the weakly compressible (WC) SPH method, and the flexible substrate is modeled by the total Lagrangian (TL) SPH method. Liquid surface tension is modeled by the continuum surface force (CSF) method through reconstructing the free surface of the droplet. The surface geometry reconstruction process is given for both 2D and 3D cases, enabling the model to simulate the bouncing of droplets in both 2D and 3D cases. Furthermore, in order to reduce the computational cost of 3D simulation, the TL-SPH model combined with shell theory established the TL-SPH-shell model, thus achieving the simulation of 3D droplet impact on thin plates. For verification and testing, the established WC-TL SPH model is used to simulate the process of droplet impact on various substrates, including fixed-fixed and curved beams, micro-pillar substrates, and cantilever beams. The results show that the spreading, retraction and bouncing processes of the droplets interact with the deformation and recovery of the flexible substrate, accompanied by large deformation and dynamic characteristics, which can be effectively simulated by the model.

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1. Introduction

Droplet impact is a common phenomenon that has attracted extensive attention and has been a research hot spot. Most previous studies have assumed that the substrate is rigid; however, in practical applications, the surface impacted by droplets cannot be ideally equivalent to a rigid substrate. Flexible or soft substrates (Mangili et al., 2012; Howland et al., 2016) and the deformation of liquid films (Chen et al., 2017) also play an important role. Examples of this include raindrops impacting on insect wings (Ha et al., 2013) and plant leaves (Gart et al., 2015), droplets impacting on the surface of energy harvesters (Helseth and Wen, 2017), and spraying insecticides on leaves (Dong et al., 2015), which can cause significant deformation of the substrate. The effect of water droplets on a flexible substrate is quite different from that on a rigid substrate. As droplets bounce or spread across the surface, the substrate deforms and even vibrates (Weisensee et al., 2017), affecting the impact mechanism. Studying the effect of droplets on flexible structures can not only deepen

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our understanding of this particular fluid–solid coupling process, but also provide insight into the design of new materials and sensors.

In the field of surface bionics, it is common practice to improve surface hydrophobicity by preparing substrates with different micro-structures or adding additives to the droplets (Aria and Gharib, 2014; Bergeron et al., 2000). Previous work showed that the substrate elasticity can enhance the surface hydrophobicity through a different mechanism (Vasileiou et al., 2016). By adjusting parameters such as stiffness and droplet diameter, the contact time can be significantly reduced. Gart et al. (2015) compared the response characteristics of hydrophobic and hydrophilic micro-beams under droplet impact, and established an analytical model to predict the deformation and vibration of the beam. Weisensee et al. (2017) studied the mechanical mechanism of droplet impact on a super-hydrophobic surface and found that the contact time of the droplet on the surface was significantly reduced due to the elasticity of the surface. The springboard effect of the elastic surface can produce the pancake bouncing of the droplet. Researchers had established the contact time model and beam vibration model to describe the process of a single droplet impact on a simple beam (Huang et al., 2019; Kim et al., 2018; Chen et al., 2019). However, for flexible substrates with more complex structures, it is hard to establish such an analytical model. Indeed, a droplet impacts on a deformed surface can be seen as a special process of fluid–solid coupling (Dong et al., 2019), which is closely related to the properties of the droplet (diameter, density, velocity, surface tension) and the impacted surface (roughness, elastic parameters, stiffness). In consideration of the difficulty of establishing an analytical model or precisely controlling influential factors in the experiment, numerical simulation may provide an effective solution for this problem.

Previous numerical models of droplet impact have mostly been established using computational fluid dynamics (CFD) methods, where the substrate is usually assumed to be rigid. Traditional mesh-based CFD methods, such as fluid volume method (Guo et al., 2014), level set method (Tanguy and Berlemont, 2005), front tracking method (Muradoglu and Tasoglu, 2010), use a fixed Euler grid to mesh the computational domain, where both the droplet and the ambient fluid (such as air) are considered. Simulation of droplet impact must be carried out in a multiphase flow model, and the fluid interface needs to be tracked and constructed during the simulation (Tanguy and Berlemont, 2005). Different methods of coupling need to be employed to solve the fluid–solid coupling problem. Complex coupling and interface treatment algorithms may lead to low computational efficiency and poor convergence (Dong et al., 2019). Droplet impact processes involve nonlinear behaviors caused by spreading and retraction. Severe splashing may occur at a high Weber number. The accuracy and stability of the process depend on the continuity of the fluid interface (Ma et al., 2021). Droplet impact may cause a large deflection of the flexible substrate, making it difficult to track the liquid–solid interface and free surface of the droplet.

The mesh-less method uses discrete particles instead of grids and elements to discretize the computational domain. As representative mesh-less methods, molecular dynamics (MD), dissipative particle dynamics (DPD), moving particle semi-implicit (MPS), smoothed particle hydrodynamics (SPH), have been applied to simulate droplet impact processes at different scales. DPD and MD were commonly used to simulate micron-droplets (Wang et al., 2017) and nano-droplets (Gao et al., 2018) impacting the micro- and nanostructures, helping to explain droplet impact mechanisms at the micro- and nano-scales. However, physical variables such as pressure, and viscosity, which are important for droplet impact processes, are difficult to be accurately described by DPD and MD. Moreover, if MD or DPD method were used to model millimeter-scale droplets, numerous particles would be required to represent the domain, which would be computationally expensive. In contrast, MPS (Liu et al., 2022) and SPH (Ma et al., 2021) were developed based on the theory of continuum mechanics. Combined with the surface tension model, MPS and SPH methods can also be applied to simulate capillary-scale problems (i.e., surface-tension-dominated flows) (Ma et al., 2021; Liu et al., 2022). The SPH method is suitable for the simulation of complex flows (Dong et al., 2022), provided that many variants and improved algorithms can ensure the accuracy and stability of the 3D simulation (Liu et al., 2017; Chen et al., 1999; Huang et al., 2015). Moreover, elastic constitutive models and fluid–solid coupling algorithms can be used to simulate the interaction between the droplet and the solid surface (Wang et al., 2021). In addition, the Lagrange nature of SPH makes it advantageous in large deformation problems (Liu and Liu, 2003).

The SPH method is adopted to establish a fluid–solid coupling model for simulation of droplet impact on a flexible substrate. In SPH, the fluid–solid coupling can be realized in two ways: the first is that the fluid is described by SPH, while the solid is modeled by other methods, such as SPH-FEM coupling (Yang et al., 2012); the second is that both the fluid and solid are described by SPH (Antoci et al., 2007; Liu et al., 2013). Antoci et al. (2007) first established a fully SPH fluid–solid coupling model for hydro-elastic problems, where the fluid was modeled by weakly compressible SPH and the solid was modeled by elastic SPH (Rafiee and Thiagarajan, 2009). Abbas et al. (2018) used the Poisson method to establish a fluid model to improve the pressure evaluation of the fluid–solid interface. The standard SPH method based on updated Lagrangian formulation suffers from tension instabilities when solving solid mechanics (Sweble et al., 1995). In the tensile stress state, SPH particles that describe solid materials are prone to nonphysical clumping. To solve the problem, several approaches have been proposed, including artificial stress or pressure method (Monaghan, 2000; Gray et al., 2001), conservative smoothing approach (Randles and Libersky, 1996; Sweble et al., 1994; Balsara, 1995). It is commonly believed that tension instabilities arise from the use of continuously updated particle positions to compute the Euler kernel (Belytschko et al., 2000). A stability analysis showed that Lagrangian kernel functions do not suffer from this problem (Rabczuk et al., 2004). Hence, the total Lagrangian (TL) SPH formulation was proposed to solve the tension instability. Using the Lagrangian kernel to reformulate solid equilibrium equations, the TL-SPH method demonstrates good stability and accuracy. In the TL-SPH method, the equilibrium equations are established on the reference configuration. The kernel

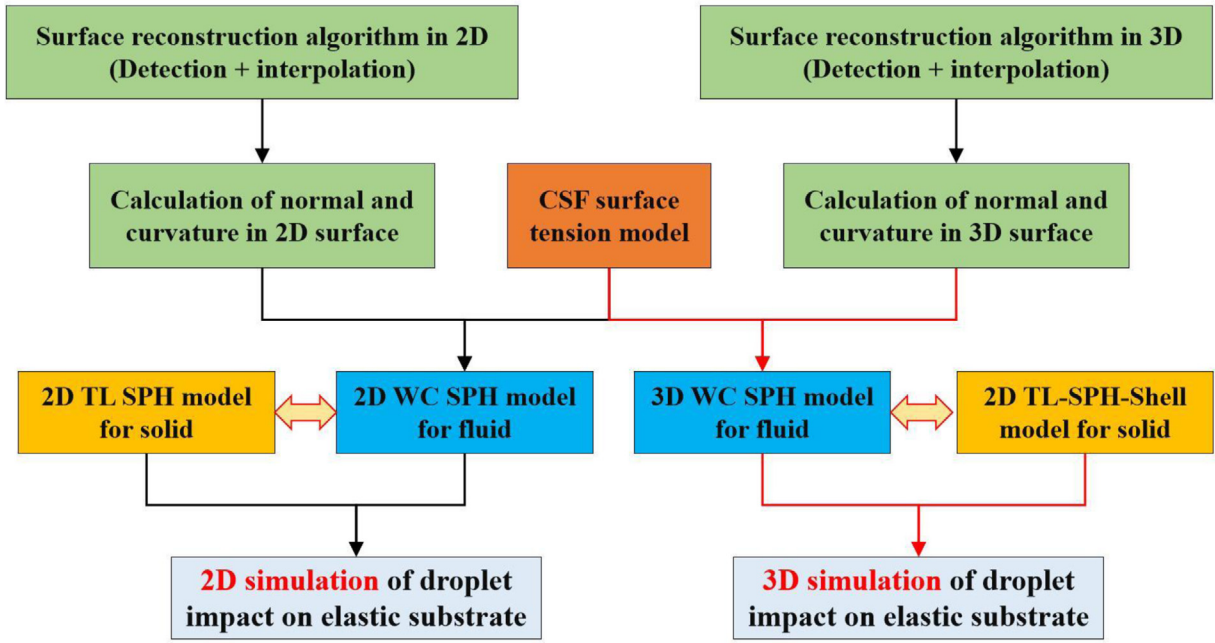


Fig. 1. Modeling strategy for simulating droplet impact on flexible substrates.

and kernel gradient are calculated only once based on the reference configuration, thereby avoiding the time-consuming process of searching for neighboring particles (Wang et al., 2021).

The rest of the paper is organized as follows. In Section 2, the basic theory of the kernel and particle approximations is introduced, as well as the theoretical background of weakly compressible (WC) SPH formulation. The modeling procedure of surface tension is given for both 2D and 3D cases. In Section 3, the total Lagrangian (TL) SPH formulation is introduced, along with details of algorithms of fluid–solid coupling model and a numerical flowchart for implementation. In Section 4, the TL-SPH shell model is established for thin structures. 3D simulation of a droplet impact on a cantilever can be simulated. In Section 5, the proposed SPH models are used to simulate droplet impacts on various types of flexible substrates. Simulation results are analyzed and discussed, and the effects of different parameters on the droplet impact are studied. Fig. 1 shows the modeling strategy adopted in this study.

2. Weakly compressible (WC) SPH model

2.1. Kernel and particle approximations

The computational domain is discretized into a number of particles which carry material properties and field variables. Based on these discrete particles, the governing equation can be transformed into discrete equations. In the SPH method, the derivation of the discrete equations has two steps in terms of the kernel and particle approximations. The kernel approximation of the field function $f(\mathbf{x})$ is expressed as:

$$f(\mathbf{x}) = \int_{\Omega_{\mathbf{x}}} f(\mathbf{x}') W(r, h) d\mathbf{x}', \quad (1)$$

where $\Omega_{\mathbf{x}}$ denotes the support domain, $W(r, h)$ is the kernel function, r is the radius, $r = |\mathbf{x} - \mathbf{x}'|$, and h is the smoothing length.

As shown in Fig. 2, the kernel $W(r, h)$ is a monotonically decreasing symmetric function. It has the following properties: normalization, compactness, non-negativity, monotonicity, Dirac function. The normalization condition is defined as:

$$\int_{\Omega_{\mathbf{x}}} W(|\mathbf{x} - \mathbf{x}'|, h) d\mathbf{x}' = 1, \quad (2)$$

The compactness condition determines the boundary of the support domain: if $|\mathbf{x} - \mathbf{x}'| > kh$, $W(\mathbf{x} - \mathbf{x}', h) = 0$, where k is the proportional factor, and the product kh represents the maximum radius of the support domain. The cubic spline

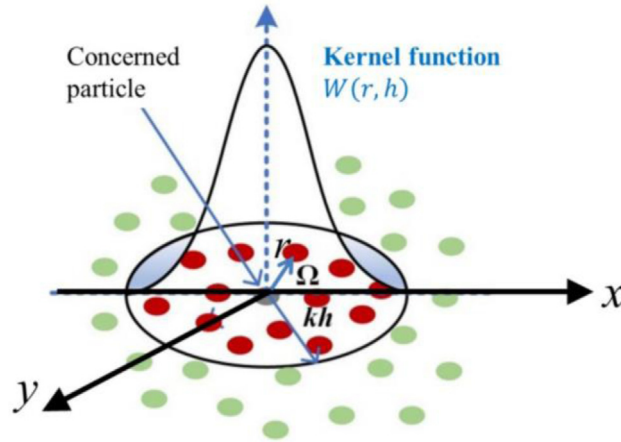


Fig. 2. Kernel function and support domain.

function is selected as the kernel function, and it is expressed as:

$$W(q, h) = \alpha_d \times \begin{cases} \frac{2}{3} - q^2 + \frac{1}{2}q^3, & 0 \leq q \leq 1 \\ \frac{1}{6}(2 - q)^3, & 1 \leq q \leq 2 \\ 0, & q \geq 2, \end{cases} \quad (3)$$

where q is the non-dimensional radius, $q = r/h$, and the parameter $\alpha_d = 15/7\pi h^2$ for 2D, $\alpha_d = 3/2\pi h^2$ for 3D. In SPH, interacting particles always appear in pairs. The kernel function of interacting particles i and j is written as $W(|\mathbf{x}_i - \mathbf{x}_j|, h_{ij})$ or $W(|\mathbf{x}_{ij}|, h_{ij})$, where h_{ij} denotes the mean value of h_i and h_j , $h_{ij} = \frac{h_i + h_j}{2}$. In this study, the smoothing length is constant, and the solid and liquid phases have the same smoothing length.

The kernel approximation of the derivative of the function $f(\mathbf{x})$ is written as:

$$\nabla f(\mathbf{x}) = \int_{\Omega_{\mathbf{x}}} f(\mathbf{x}') \nabla_{\mathbf{x}} W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}', \quad (4)$$

where $\nabla_{\mathbf{x}} W(\mathbf{x} - \mathbf{x}', h)$ denotes the kernel gradient.

The particle approximation is the discretized form of kernel approximation. The particle approximation of a field function and its derivative is written as:

$$\begin{cases} \langle f(\mathbf{x}_i) \rangle = \sum_{j=1}^{N_i} f(\mathbf{x}_j) W(\mathbf{x}_i - \mathbf{x}_j, h) V_j \\ \langle \nabla f(\mathbf{x}_i) \rangle = \sum_{j=1}^{N_i} f(\mathbf{x}_j) \nabla_i W(\mathbf{x}_i - \mathbf{x}_j, h) V_j \end{cases}, \quad (5)$$

where i is the index of the concern particle, and particle j is in the support domain of particle i . V_j is the volume, $V_j = m_j/\rho_j$, where m_j is the mass, ρ_j is the density. N_i is the number of particles located in the support domain of particle i .

2.2. WC-SPH formulation

The Navies-Stokes equation is used as the governing equation for the fluid phase (i.e., the liquid droplet). The equation of state (EOS) with an artificial sound speed is adopted to close the governing equations. Through properly selecting the value of sound speed, the variation of material density can be limited in a small value. The SPH method based on this solution strategy is called weakly compressible (WC) SPH method (Becker and Teschner, 2007). The WC-SPH method has been widely used to simulate free surface flows (Monaghan, 1994). However, the drawback of WC-SPH is the unphysical oscillations in the evaluation of pressure. Many modifications have been proposed to reduce these fluctuations. The introduction of numerical diffusive terms such as artificial viscosity and density diffuse term can provide stable solution even for violent flows, such as wave breaking, liquid flow impact (Antuono et al., 2012; Shadloo et al., 2012). The corresponding scheme to WC-SPH is the Incompressible Smoothed Particle Hydrodynamics (ISPH) scheme. The ISPH scheme requires the solution of a Poisson equation, which results in an implicit algorithm of pressure evaluation. It is

more complex to implement than WC-SPH and requires more computational effort, even though it uses larger time steps (Daly et al., 2016).

The governing equations, including the continuity and momentum equations, are written as:

$$\frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v}, \quad (6)$$

$$\frac{d\mathbf{v}}{dt} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} + \mathbf{f}, \quad (7)$$

where t denotes the time, ρ is the material density, $\mathbf{v}$ is the velocity, and $\mathbf{f}$ denotes the external force. p represents the pressure. ν represents the kinematic viscosity.

Based on the particle approximation Eq. (5), combined with appropriate mathematical transformations, the terms on the right-hand side of Eqs. (6) and (7) can be written in the particle discrete form as:

$$(\nabla \cdot \mathbf{v})_i = \frac{1}{\rho_i} \sum_{j=1}^{N_i} m_j \mathbf{v}_{ji} \cdot \nabla_i W(\mathbf{x}_{ij}, h_{ij}), \quad (8)$$

$$\left(\frac{1}{\rho} \nabla p\right)_i = \sum_{j=1}^{N_i} m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2}\right) \nabla_i W(\mathbf{x}_{ij}, h_{ij}), \quad (9)$$

$$(\nu \nabla^2 \mathbf{v})_i = \sum_{j=1}^{N_i} \frac{4m_j (\mu_i + \mu_j)}{(\rho_i + \rho_j)^2} \frac{\mathbf{x}_{ij} \cdot \nabla_i W(\mathbf{x}_{ij}, h_{ij})}{|\mathbf{x}_{ij}|^2 + (0.01h_{ij})^2} \mathbf{v}_{ij}. \quad (10)$$

The SPH discrete equations of governing equations can be written as:

$$\frac{d\rho_i}{dt} = \sum_{j=1}^{N_i} m_j (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla_i W(\mathbf{x}_i - \mathbf{x}_j, h), \quad (11)$$

$$\frac{d\mathbf{v}_i}{dt} = -\sum_{j=1}^{N_i} m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2}\right) \nabla_i W(\mathbf{x}_{ij}, h) + \sum_{j=1}^{N_i} \frac{4m_j (\mu_i + \mu_j)}{(\rho_i + \rho_j)^2} \frac{\mathbf{x}_{ij} \cdot \nabla_i W(\mathbf{x}_i - \mathbf{x}_j, h)}{|\mathbf{x}_{ij}|^2 + (0.01h)^2} \mathbf{v}_{ij} + \mathbf{f}_i, \quad (12)$$

where $\mathbf{x}_{ij} = \mathbf{x}_i - \mathbf{x}_j$.

The pressure is computed by the following EOS (Monaghan, 1994):

$$p = \frac{c^2 \rho_0}{\gamma} \left(\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right), \quad (13)$$

where the constant $\gamma = 7.0$ (Monaghan, 1994). c is sound speed taken as 10 times the maximum velocity, $c = 10V_{max}$. ρ_0 is the reference density of the fluid phase.

The artificial viscous force is introduced in the momentum equation to stabilize the simulation. The term of artificial viscosity is written as:

$$\Pi_i^{art} = -\sum_{j=1}^{N_i} m_j \pi_{ij}^{art} \nabla_i W(\mathbf{x}_i - \mathbf{x}_j, h), \quad (14)$$

where π_{ij}^{art} is expressed as :

$$\pi_{ij}^{art} = \begin{cases} -\alpha \frac{c \varphi_{ij}}{\bar{\rho}_{ij}} & \mathbf{v}_{ij} \cdot \mathbf{x}_{ij} < 0 \\ 0 & \mathbf{v}_{ij} \cdot \mathbf{x}_{ij} \geq 0 \end{cases}, \quad (15)$$

where $\bar{\rho}_{ij} = \frac{\bar{\rho}_i + \bar{\rho}_j}{2}$, $\varphi_{ij} = h_{ij} \frac{\mathbf{v}_{ij} \cdot \mathbf{x}_{ij}}{|\mathbf{x}_{ij}|^2 + 0.01h^2}$, $\alpha = 0.075$.

2.3. Surface tension modeling

Approaches for modeling surface tension in SPH are mainly divided in two classes (Breinlinger et al., 2013a). The first class adopts the continuous surface force (CSF) method (Brackbill et al., 1992) by applying a volumetric force on the surface or interface of the fluid. The second class introduces an explicit particle–particle interaction force (PIF) to mimic cohesive forces in molecular dynamics (Tartakovsky and Meakin, 2005). Morris (2000) first introduced the CSF method in the SPH method. The surface tension force is expressed as the function of interface curvature and normal vector, which can be computed using color function (Morris, 2000; Li et al., 2018) or through geometry reconstruction (Zhang et al.,

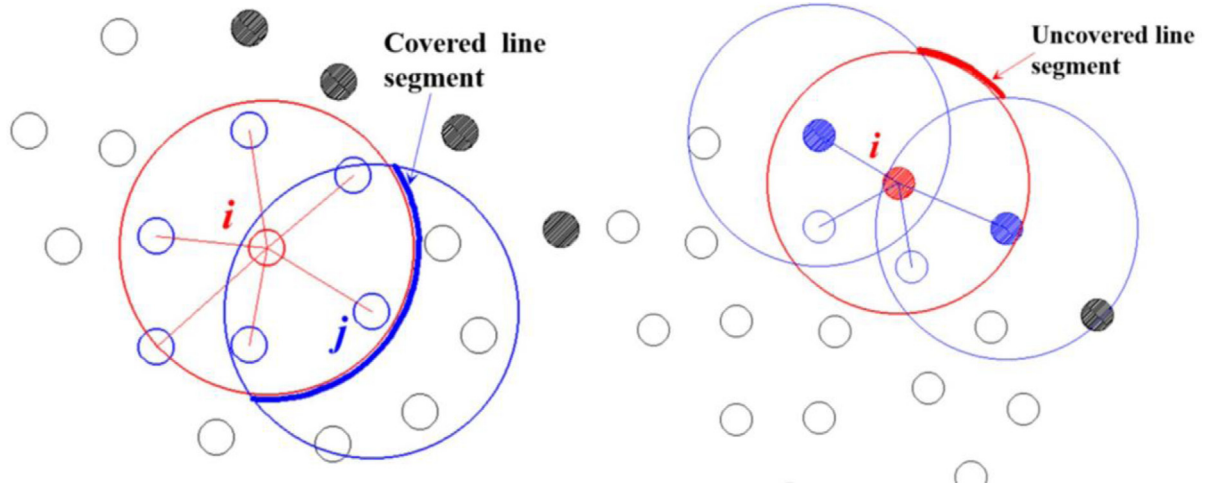


Fig. 3. Schematic diagram of boundary particle detection in 2D.

2012). The advantage of CSF is that the model parameters are consistent with physical reality, which is convenient for practical applications.

In this study, the CSF method is used to simulate the surface tension of liquid droplets. Since the droplet has a free surface boundary, the geometric reconstruction of the free surface is required in order to calculate the curvature and normal vector of the droplet surface. Following the computational procedure proposed by Dils (2000) and Zhang et al. (2012), boundary particles on the surface of the droplet are detected first.

As shown in Fig. 3, the boundary particle detection is realized by scanning the radius of the support domain around a SPH particle. If the circle of the SPH particle is not completely covered by the circles of its neighbors, this particle is labeled as a boundary particle. Using this method, all boundary particles of the droplet can be detected and the surface tension calculated by the CSF model is applied to these particles. After determining the surface boundary particles, these surface particles can be treated as discrete points and the geometric surface can be reconstructed based on these points by interpolation methods. In the following, the adopted surface detection and interpolation methods will be described for the 2D and 3D cases, respectively.

According to the CSF model, the surface tension force is expressed as:

$$\mathbf{f} = \delta \kappa \mathbf{n} \lambda, \quad (16)$$

where δ denotes the coefficient of surface tension, N/m; κ denotes the curvature of the boundary surface, and $\mathbf{n}$ is the unit normal vector of the surface. The parameter λ is set to a constant $1/\Delta x$ (Zhang et al., 2012), where Δx is the initial particle spacing.

2.3.1. For two-dimensional case

In 2D, the boundary particles are treated as a series of discrete points on the surface of the droplet, through which the surface curve is reconstructed by Lagrangian interpolation method. Firstly, the global coordinate system is converted into a local coordinate system, as shown in Fig. 4, and for any boundary particle i , the position vector of the origin O' of the local coordinate system is determined according to the following equation:

$$\mathbf{x}_{O'} = \sum_j \mathbf{x}_j / N_j, \quad (17)$$

where j represents the domain particles of the boundary particle i , N_j is the number of domain particles, and $\mathbf{x}_j$ is the position vector of the particle j . Note that the direction of the local coordinate system axis y' should be pointed from the origin O' to particle i .

The relationship between the local and global coordinate systems is as follows:

$$\begin{cases} x' = x \cos \alpha + y \sin \alpha \\ y' = -x \sin \alpha + y \cos \alpha \end{cases}, \quad (18)$$

where α is the angle between the x -axis of the global coordinate system and the x' axis of the local coordinate system. Subsequently, with all boundary particles in the support domain of boundary particle i as interpolation points, the

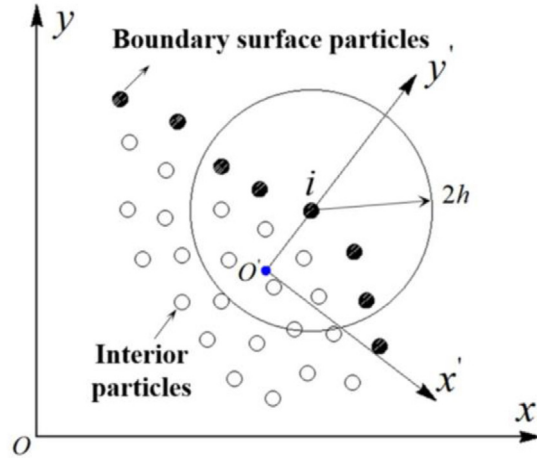


Fig. 4. The local coordinate system of boundary particle i .

interpolation polynomial is obtained as:

$$\begin{cases} P(x) = \sum_j P_j(x) \\ P_j(x) = y_j \prod_{k \neq j} \frac{(x - x_k)}{(x_j - x_k)} \end{cases} \quad (19)$$

where j is the boundary particle in the support domain of i . y_j is the y -coordinate of particle j (local coordinate system), and x_j is the x -coordinate of particle j (local coordinate system).

Then, the curvature at the surface point is calculated by the equation:

$$\kappa_i = \frac{|P''(x_i)|}{(1 + P'^2(x_i))^{3/2}}. \quad (20)$$

Similarly, the normal vector at any boundary particle i is calculated by the following equation (Zhang et al., 2012):

$$\mathbf{n}_i = \begin{cases} \langle P'(x_i), -1 \rangle & \text{if } P''(x_i) < 0 \\ \langle -P'(x_i), 1 \rangle & \text{if } P''(x_i) > 0 \end{cases}. \quad (21)$$

Fig. 5 shows the simulation results of a droplet impacting a solid wall. The direction of the arrow indicates the direction of the normal vector, which is the direction of the surface tension. As the figure shows, surface tension always tends to flatten the surface.

The calculated normal vector needs to be converted to a global coordinate system by the following equation (Zhang et al., 2012):

$$\mathbf{n}_i = \begin{cases} \langle P'(x_i) \cos \alpha + \sin \alpha, P'(x_i) \sin \alpha - \cos \alpha \rangle & \text{if } P''(x_i) < 0 \\ \langle -P'(x_i) \cos \alpha - \sin \alpha, -P'(x_i) \sin \alpha + \cos \alpha \rangle & \text{if } P''(x_i) > 0 \end{cases}. \quad (22)$$

2.3.2. For three-dimensional case

For 3D simulations, a similar surface reconstruction strategy is adopted. The boundary particle detection and interpolation methods are more complex than those of 2D case. Boundary particles are detected using the algorithm developed by Haque and Dils (2007), and then the boundary surface is reconstructed based on the moving least squares (MLS) method (Dils, 2000). Before reconstructing the boundary surface, the coordinate system is converted to the local coordinate system. The local coordinate origin o' for the boundary particle i is calculated as the average point of its neighbors. The direction vectors of the coordinate axes (i.e., $\langle x', y', z' \rangle$) in the local coordinate system can be expressed as:

$$\begin{cases} \hat{\mathbf{z}}_{io} = \frac{(\mathbf{x}_i - \mathbf{x}_{o'})}{|\mathbf{x}_i - \mathbf{x}_{o'}|} \\ \hat{\mathbf{x}}_{io} = \frac{\hat{\mathbf{x}} \times \hat{\mathbf{z}}_{io}}{|\hat{\mathbf{x}} \times \hat{\mathbf{z}}_{io}|} \\ \hat{\mathbf{y}}_{io} = \frac{\hat{\mathbf{z}}_{io} \times \hat{\mathbf{x}}_{io}}{|\hat{\mathbf{z}}_{io} \times \hat{\mathbf{x}}_{io}|} \end{cases}, \quad (23)$$

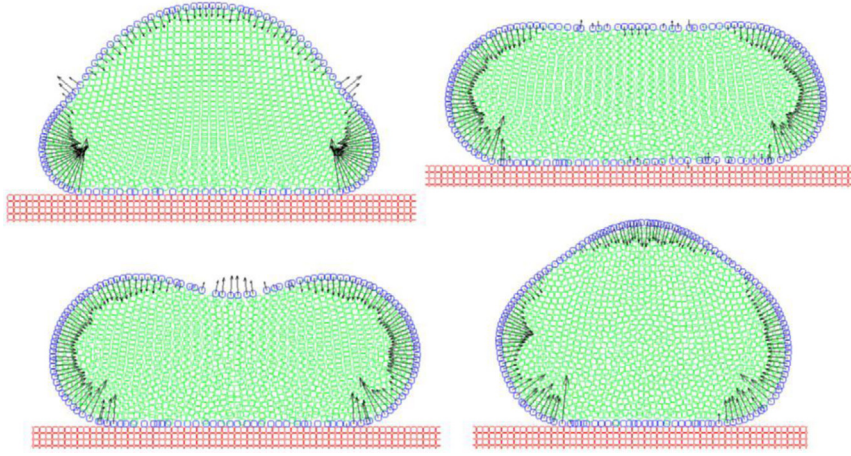


Fig. 5. Simulation of droplet impacts on a solid surface, showing the distribution of normal vector of the surface.

where $\hat{\mathbf{x}}_{io}$, $\hat{\mathbf{y}}_{io}$, and $\hat{\mathbf{z}}_{io}$ are the unit vectors of the local coordinate system axes in the global coordinate system. $\mathbf{x}_i$ is the position vector of boundary particle i , and $\mathbf{x}_o$ is the position vector of the origin of the local coordinate.

The following conversion relation exists between local and global coordinate systems:

$$\begin{cases} x' = \mathbf{x} \cdot \hat{\mathbf{x}}_{io} \\ y' = \mathbf{x} \cdot \hat{\mathbf{y}}_{io} \\ z' = \mathbf{x} \cdot \hat{\mathbf{z}}_{io} \end{cases} \quad (24)$$

Then, the boundary surface is reconstructed in the transformed coordinate system using the MLS method, which is used to reconstruct a three-dimensional surface from a set of points. To obtain the second order derivative of the surface function, a quadratic basis is used:

$$\mathbf{p}^T = [1, x, y, x^2, xy, y^2]. \quad (25)$$

The surface function can be approximated as:

$$\begin{cases} z^h(\mathbf{x}) = \sum_{j=1}^m p_j(\mathbf{x}) a_j(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{a}(\mathbf{x}) \\ \mathbf{a}^T(\mathbf{x}) = [a_1(\mathbf{x}), a_2(\mathbf{x}), \dots, a_m(\mathbf{x})] \end{cases} \quad (26)$$

where the coefficient vector $\mathbf{a}(\mathbf{x})$ is a function of $\mathbf{x}$, and it can be obtained by minimizing the following weighted discrete L_2 norm:

$$J_2 = \sum_{i=1}^n \hat{W}(\mathbf{x} - \mathbf{x}_i) (\mathbf{p}^T(\mathbf{x}_i) \mathbf{a}(\mathbf{x}) - z_i)^2. \quad (27)$$

where n is the number of points included in the support domain of the weight function $\hat{W}(\mathbf{x} - \mathbf{x}_i) \neq 0$ and z_i is the value of z at $\mathbf{x} = \mathbf{x}_i$. Find the stagnation point of J_2 with respect to $\mathbf{a}(\mathbf{x})$:

$$\frac{\partial J_2}{\partial \mathbf{a}} = 0, \quad (28)$$

then the following linear relationship is obtained:

$$\mathbf{A}(\mathbf{x}) \mathbf{a}(\mathbf{x}) = \mathbf{B}(\mathbf{x}) \mathbf{Z}, \quad (29)$$

where $\mathbf{Z} = [z_1, z_2, \dots, z_n]$. The matrix $\mathbf{A}(\mathbf{x})$ is expressed as:

$$\mathbf{A}(\mathbf{x}) = \sum_{i=1}^n \hat{W}(\mathbf{x} - \mathbf{x}_i) \mathbf{p}(\mathbf{x}_i) \mathbf{p}^T(\mathbf{x}_i), \quad (30)$$

and the matrix $\mathbf{B}(\mathbf{x})$ is expressed as:

$$\mathbf{B}(\mathbf{x}) = [\hat{W}_1(\mathbf{x}) \mathbf{p}(\mathbf{x}_1), \hat{W}_2(\mathbf{x}) \mathbf{p}(\mathbf{x}_2), \dots, \hat{W}_n(\mathbf{x}) \mathbf{p}(\mathbf{x}_n)]. \quad (31)$$

According to Eq. (29), the vector $\mathbf{a}(\mathbf{x})$ is computed as:

$$\mathbf{a}(\mathbf{x}) = \mathbf{A}^{-1}(\mathbf{x}) \mathbf{B}(\mathbf{x}) \mathbf{Z}. \quad (32)$$

Substituting the above equation into Eq. (26) yields:

$$z^h(\mathbf{x}) = \sum_{i=1}^n \phi_i(\mathbf{x}) z_i = \Phi^T(\mathbf{x}) \mathbf{Z}, \quad (33)$$

where $\Phi^T(\mathbf{x}) = [\phi_1(\mathbf{x}) \quad \phi_2(\mathbf{x}) \quad \cdots \quad \phi_n(\mathbf{x})] = \mathbf{p}^T(\mathbf{x}) \mathbf{A}^{-1}(\mathbf{x}) \mathbf{B}(\mathbf{x})$. The shape function $\phi_i(\mathbf{x})$ of point i can be defined as:

$$\phi_i(\mathbf{x}) = \sum_{j=1}^m p_j(\mathbf{x}) (\mathbf{A}^{-1}(\mathbf{x}) \mathbf{B}(\mathbf{x}))_{ji} = \mathbf{p}^T(\mathbf{x}) (\mathbf{A}^{-1} \mathbf{B})_i. \quad (34)$$

Then, the first and second derivatives of the surface function are obtained as:

$$z_{,i}^h = \Phi_{,i}^T \mathbf{Z}, \quad z_{,ij}^h = \Phi_{,ij}^T \mathbf{Z}. \quad (35)$$

To calculate the normal vector and curvature of the surface, the surface function is written as:

$$\mathbf{r}(x, y) = \langle x, y, z \rangle = \langle x, y, f(x, y) \rangle, \quad (36)$$

where $z = f(x, y)$ is the surface function that is reconstructed by the MLS method.

The normal vector is calculated by the following equation (Zhang et al., 2012):

$$\mathbf{n} = [-f_x, -f_y, 1] \frac{1}{\sqrt{1 + f_x^2 + f_y^2}}, \quad (37)$$

where f_x and f_y denote the derivative of function $f(x, y)$ with respect to x and y .

The curvature of boundary surface can be calculated by the following equation (Zhang et al., 2012):

$$\kappa = \frac{(1 + f_x^2) f_{yy} - 2 f_x f_y f_{xy} + (1 + f_y^2) f_{xx}}{2 (1 + f_x^2 + f_y^2)^{3/2}}, \quad (38)$$

where f_{xx} , f_{xy} , and f_{yy} are the second derivative of surface function.

3. Total Lagrangian (TL) SPH model for 2D simulation

3.1. Governing equation in the TL-SPH method

In TL-SPH, the conservation equations are solved in the undeformed configuration which is also named as the reference configuration, denoted by $\mathbf{X}$. The change in field variables is used to calculate the current deformed configuration $\mathbf{x}$. For example, a point M_0 in the reference configuration moves from its initial position $\mathbf{X}$ to the current point M with the position vector $\mathbf{x}$. The motion of the point M_0 is expressed as:

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t), \quad (39)$$

where t denotes the time.

The conservation equations contain mass and momentum equations, which can be expressed as follows:

$$\rho(\mathbf{x}, t) = J(\mathbf{X}, t)^{-1} \rho_0(\mathbf{X}), \quad (40)$$

$$\frac{d\mathbf{v}}{dt} = \frac{1}{\rho_0} \nabla_0 \cdot \mathbf{P} + \mathbf{b}_0, \quad (41)$$

$$\mathbf{P} = J(\mathbf{X}, t) \sigma \mathbf{F}^{-T}, \quad (42)$$

where $\rho_0(\mathbf{X})$ is the undeformed material density, and $\rho(\mathbf{x}, t)$ is the deformed material density. $\mathbf{P}$ denotes the first Piola Kirchhoff stress, and σ denotes the Cauchy stress tensor. ∇_0 denotes the divergence operator or gradient operator with subscript 0 indicating that the values are calculated at the reference configuration $\mathbf{X}$. $\mathbf{b}_0$ denotes the contact force due to the impact of liquid droplet, or other volumetric forces (e.g. gravity). J is the Jacobian expressed as $J = \det(\mathbf{F})$, where $\mathbf{F}$ is the deformation gradient tensor calculated as:

$$\mathbf{F} = \frac{d\mathbf{x}}{d\mathbf{X}}. \quad (43)$$

The particle approximation form for the deformation gradient tensor is written as (Sun et al., 2021):

$$\begin{cases} \mathbf{F}_i = \sum_{j=1}^{N_i} (\mathbf{x}_j - \mathbf{x}_i) \otimes \mathbf{L}_{0i} \nabla_{0i} W_{0ij} \frac{m_{0j}}{\rho_{0j}} \\ \mathbf{L}_{0i} = \left[\sum_{j=1}^{N_i} (\mathbf{x}_j - \mathbf{x}_i) \otimes \nabla_{0i} W_{0ij} \frac{m_{0j}}{\rho_{0j}} \right]^{-1}, \end{cases} \quad (44)$$

where $\nabla_{0i} W_{0ij}$ denotes the kernel gradient computed at the reference configuration, $\nabla_{0i} W_{0ij} = \nabla_{0i} W(\mathbf{x}_i - \mathbf{x}_j, h) = \begin{bmatrix} \frac{\partial W_{0ij}}{\partial X} \\ \frac{\partial W_{0ij}}{\partial Y} \end{bmatrix}$. Because the mass of a SPH particle is conserved, $m_j = m_{0j}$. Here, the matrix $\mathbf{L}_{0i}$ is adopted to modify the kernel gradient to improve the accuracy of the boundary. It is calculated at the reference configuration as follows:

$$\mathbf{L}_{0i} = \begin{bmatrix} l_{xx} & l_{xy} \\ l_{yx} & l_{yy} \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^{N_i} (X_j - X_i) \frac{\partial W_{0ij}}{\partial X} \frac{m_{0j}}{\rho_{0j}} & \sum_{j=1}^{N_i} (X_j - X_i) \frac{\partial W_{0ij}}{\partial Y} \frac{m_{0j}}{\rho_{0j}} \\ \sum_{j=1}^{N_i} (Y_j - Y_i) \frac{\partial W_{0ij}}{\partial X} \frac{m_{0j}}{\rho_{0j}} & \sum_{j=1}^{N_i} (Y_j - Y_i) \frac{\partial W_{0ij}}{\partial Y} \frac{m_{0j}}{\rho_{0j}} \end{bmatrix}^{-1}. \quad (45)$$

The SPH method suffers from linear inconsistencies due to incomplete kernel support at the boundary of the domain. This means that the correct value of the derivative is not fully recovered in this region. For fluid dynamic simulations, this is usually handled by ensuring that the solid boundary (represented by the dummy or mirror particles) is thick enough so that fluid particles close to the boundary are always adequately supported (O'Connor and Rogers, 2021a). However, when modeling solid mechanics, the use of boundary (dummy or mirror) particles is not appropriate, necessitating the use of a correction matrix for kernel gradient evaluation. The modified kernel gradient is expressed as:

$$\tilde{\nabla}_{0i} W_{0ij} = \mathbf{L}_{0i} \nabla_{0i} W_{0ij}. \quad (46)$$

For 2D simulations, the modified kernel gradient is expressed as:

$$\tilde{\nabla}_{0i} W_{0ij} = \begin{bmatrix} \frac{\partial \tilde{W}_{0ij}}{\partial X} \\ \frac{\partial \tilde{W}_{0ij}}{\partial Y} \end{bmatrix} = \begin{bmatrix} l_{xx} & l_{xy} \\ l_{yx} & l_{yy} \end{bmatrix} \begin{bmatrix} \frac{\partial W_{0ij}}{\partial X} \\ \frac{\partial W_{0ij}}{\partial Y} \end{bmatrix} = \begin{bmatrix} l_{xx} \frac{\partial W_{0ij}}{\partial X} + l_{xy} \frac{\partial W_{0ij}}{\partial Y} \\ l_{yx} \frac{\partial W_{0ij}}{\partial X} + l_{yy} \frac{\partial W_{0ij}}{\partial Y} \end{bmatrix}, \quad (47)$$

Consequently, the deformation gradient tensor and its rate are calculated as:

$$\mathbf{F}_i = \begin{bmatrix} \sum_{j=1}^{N_i} (x_j - x_i) \frac{\partial \tilde{W}_{0ij}}{\partial X} V_{0j} & \sum_{j=1}^{N_i} (x_j - x_i) \frac{\partial \tilde{W}_{0ij}}{\partial Y} V_{0j} \\ \sum_{j=1}^{N_i} (y_j - y_i) \frac{\partial \tilde{W}_{0ij}}{\partial X} V_{0j} & \sum_{j=1}^{N_i} (y_j - y_i) \frac{\partial \tilde{W}_{0ij}}{\partial Y} V_{0j} \end{bmatrix}, \quad (48)$$

$$\dot{\mathbf{F}}_i = \begin{bmatrix} \sum_{j=1}^{N_i} (u_j - u_i) \frac{\partial \tilde{W}_{0ij}}{\partial X} V_{0j} & \sum_{j=1}^{N_i} (u_j - u_i) \frac{\partial \tilde{W}_{0ij}}{\partial Y} V_{0j} \\ \sum_{j=1}^{N_i} (v_j - v_i) \frac{\partial \tilde{W}_{0ij}}{\partial X} V_{0j} & \sum_{j=1}^{N_i} (v_j - v_i) \frac{\partial \tilde{W}_{0ij}}{\partial Y} V_{0j} \end{bmatrix}, \quad (49)$$

where $\mathbf{x}_i = (x_i, y_i)$ and $\mathbf{v}_i = (u_i, v_i)$ represent the position and velocity vectors of the particle, respectively. V_{0j} is the volume of the particle, $V_{0j} = \frac{m_{0j}}{\rho_{0j}}$.

The conservation equations are expressed in particle-discrete form as:

$$\rho_i = (\det(\mathbf{F}_i))^{-1} \rho_{0i}, \quad (50)$$

$$\frac{d\mathbf{v}_i}{dt} = \sum_{j=1}^{N_i} m_j \left(\frac{\mathbf{P}_i}{\rho_{0i}^2} + \frac{\mathbf{P}_j}{\rho_{0j}^2} + \boldsymbol{\Pi}_{ij} \right) \cdot \tilde{\nabla}_{0i} W_{0ij}, \quad (51)$$

where $\mathbf{P}_i = \det(\mathbf{F}_i)\sigma_i\mathbf{F}_i^{-T}$, $\mathbf{P}_j = \det(\mathbf{F}_j)\sigma_j\mathbf{F}_j^{-T}$. $\mathbf{II}_{ij}$ is the artificial viscosity which is expressed as:

$$\begin{cases} \mathbf{II}_{ij} = \det(\mathbf{F}_i)\pi_{ij}\mathbf{F}_i^{-T} \\ \pi_{ij} = \frac{\alpha ch}{\rho_0} \frac{(\mathbf{v}_j - \mathbf{v}_i) \cdot (\mathbf{x}_j - \mathbf{x}_i)}{|\mathbf{x}_j - \mathbf{x}_i|^2 + (0.01h)^2}, \end{cases} \quad (52)$$

where α is the constant, and is set to 0.2. ρ_0 is the undeformed material density of the solid. $\mathbf{v}_i$ and $\mathbf{v}_j$ are the velocities of particles i and j approaching each other.

It can be seen that the density of the deformed material can be directly calculated from the deformation gradient tensor and the undeformed density. Note that the direct use of modified kernel gradients instead of kernel gradients in the momentum equation leads to the non-conservation of linear momentum. To solve this problem, a feasible solution is to adopt the respective modified kernel gradients for particle pairs i and j , respectively, and then sum them up, as adopted in the literature (O'Connor and Rogers, 2021b). However, this treatment is performed at the cost of no longer satisfying strict linear consistency. Preliminary tests shown that the present approach (consistent but non-conservative) is stable, confirming the approach adopted in this work.

3.2. Constitutive model

The Cauchy stress $\boldsymbol{\sigma}$ tensor is decomposed into an isotropic part and a deviatoric part as follows:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}, \quad (53)$$

where p is the hydrostatic pressure, and $\boldsymbol{\tau}$ is the shear stress tensor. The stress rate is calculated as (Islam and Peng, 2019):

$$\dot{\boldsymbol{\tau}} = 2G \left(\boldsymbol{\varepsilon} - \frac{1}{3} \text{tr}(\boldsymbol{\varepsilon})\mathbf{I} \right) + \boldsymbol{\tau}\boldsymbol{\omega} - \boldsymbol{\omega}\boldsymbol{\tau}, \quad (54)$$

where G is the shear modulus, $\boldsymbol{\varepsilon}$ denotes the strain rate tensor, and $\boldsymbol{\omega}$ denotes the rotation rate tensor.

The strain rate and rotation rate tensors can be computed from the deformation gradient tensor as:

$$\boldsymbol{\varepsilon} = \frac{1}{2} (\dot{\mathbf{F}}\mathbf{F}^{-1} + \mathbf{F}^{-T}\dot{\mathbf{F}}^{-T}), \quad (55)$$

$$\boldsymbol{\omega} = \frac{1}{2} (\dot{\mathbf{F}}\mathbf{F}^{-1} - \mathbf{F}^{-T}\dot{\mathbf{F}}^{-T}), \quad (56)$$

For two dimensional and plane strain problems, the complete expression for the stress rate is:

$$\begin{cases} \frac{d\tau_{xx}}{dt} = 2G \left(\boldsymbol{\varepsilon} - \frac{1}{3} \text{tr}(\boldsymbol{\varepsilon})\mathbf{I} \right) + \boldsymbol{\tau}\boldsymbol{\omega} - \boldsymbol{\omega}\boldsymbol{\tau}, \\ \frac{d\tau_{yy}}{dt} = 2G \left(\boldsymbol{\varepsilon} - \frac{1}{3} \text{tr}(\boldsymbol{\varepsilon})\mathbf{I} \right) + \boldsymbol{\tau}\boldsymbol{\omega} - \boldsymbol{\omega}\boldsymbol{\tau}, \\ \frac{d\tau_{xy}}{dt} = \frac{d\tau_{yx}}{dt} = 2G\boldsymbol{\varepsilon} + \boldsymbol{\tau}\boldsymbol{\omega} - \boldsymbol{\omega}\boldsymbol{\tau} \\ \frac{d\tau_{zz}}{dt} = \end{cases} \quad (57)$$

The hydrostatic pressure is calculated using the equation of state as:

$$p = K \left(\frac{\rho}{\rho_0} - 1 \right), \quad (58)$$

where K denotes the bulk modulus, $K = \frac{E}{3(1-2\nu)}$, where E is the Young's modulus, and ν is the Poisson's ratio. The first Piola Kirchhoff stress can be written as:

$$\mathbf{P} = \mathbf{J}\boldsymbol{\sigma}\mathbf{F}^{-T} = \mathbf{J}(-p\mathbf{I} + \boldsymbol{\tau})\mathbf{F}^{-T} = \mathbf{J}\boldsymbol{\tau}\mathbf{F}^{-T} - \mathbf{J}p\mathbf{F}^{-T}. \quad (59)$$

3.3. Time integration scheme

The explicit time integration scheme is used to solve the fluid–solid coupling SPH model. The WC-SPH and TL-SPH models are solved in the same time integration loop. We adopt a prediction–correction scheme to update variables. This scheme predicts the evolution in time as:

$$\mathbf{v}_i^{n+1/2} = \mathbf{v}_i^n + \frac{\Delta t}{2} \dot{\mathbf{v}}_i^n, \mathbf{x}_i^{n+1/2} = \mathbf{x}_i^n + \frac{\Delta t}{2} \dot{\mathbf{x}}_i^n, \boldsymbol{\tau}_i^{n+1/2} = \boldsymbol{\tau}_i^n + \frac{\Delta t}{2} \dot{\boldsymbol{\tau}}_i^n \quad (60)$$

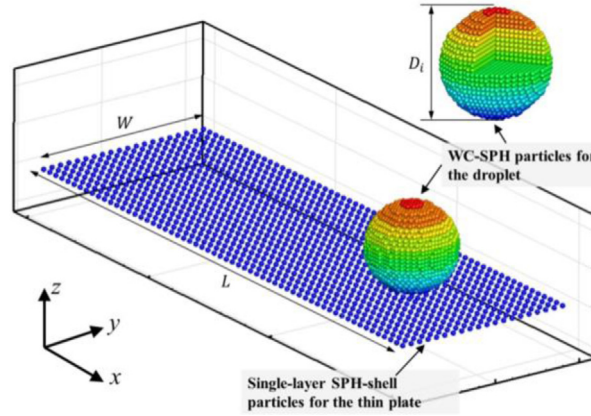


Fig. 6. SPH model of a 3D droplet impact on an elastic plate. The thin plate substrate is description by single-layer shell particles.

where $\dot{\mathbf{v}}_i^n$ is calculated by Eq. (51), and $\dot{\boldsymbol{\tau}}_i^n$ is calculated by Eq. (54). The change rate of position vector is calculated by the following equation:

$$\dot{\mathbf{x}}_i^n = \mathbf{v}_i^n + \delta \mathbf{x}_i^{XSPH}, \quad (61)$$

where $\delta \mathbf{x}_i^{XSPH}$ denotes the position correction vector computed by XSPH algorithm as:

$$\delta \mathbf{x}_i^{XSPH} = \epsilon \sum_j \frac{m_j}{\rho_{ij}} (\mathbf{v}_j - \mathbf{v}_i) W_{0ij}. \quad (62)$$

These values are then corrected using variables at the half step:

$$\mathbf{v}_i^{n+1/2} = \mathbf{v}_i^n + \frac{\Delta t}{2} \dot{\mathbf{v}}_i^{n+1/2}, \quad \mathbf{x}_i^{n+1/2} = \mathbf{x}_i^n + \frac{\Delta t}{2} \dot{\mathbf{x}}_i^{n+1/2}, \quad \boldsymbol{\tau}_i^{n+1/2} = \boldsymbol{\tau}_i^n + \frac{\Delta t}{2} \dot{\boldsymbol{\tau}}_i^{n+1/2}. \quad (63)$$

Finally, the values are calculated at the end of the time step following:

$$\mathbf{v}_i^{n+1} = 2\mathbf{v}_i^{n+1/2} - \mathbf{v}_i^n, \quad \mathbf{x}_i^{n+1} = 2\mathbf{x}_i^{n+1/2} - \mathbf{x}_i^n, \quad \boldsymbol{\tau}_i^{n+1} = 2\boldsymbol{\tau}_i^{n+1/2} - \boldsymbol{\tau}_i^n. \quad (64)$$

4. TL-SPH shell model for 3D simulation

When the TL-SPH method is used to model a 3D slender structure, such as a thin plate, several layers of particles need to be distributed in the thickness direction to capture the bending effect, which will greatly reduce the efficiency. To improve the efficiency of TL-SPH method, a shell model based on TL-SPH formulation is established. [Maurel and Combescure \(2008\)](#) developed a SPH shell model based on the Mindlin–Reissner theory. The model uses a single layer of SPH particles to discretize the plate-shell structure, which is used to predict the dynamic elastic–plastic and fracture behavior of shells. [Ming et al. \(2013\)](#) proposed a SPH shell model that uses TL-SPH formulation as the governing equation and MLS function as shape function, towards improving stability and consistency. [Zhang et al. \(2014\)](#) proposed several treatments including hourglass model, boundary conditions, irregular structures for the practical applications of TL-SPH shell model solving large-deformation and other nonlinear problems.

Based on the Reissner–Mindlin shell theory, the TL-SPH shell formulation also showed good accuracy for modeling thick plate when comparing with robust FEM method ([Lin et al., 2014](#)). In summary, using the TL-SPH shell formulation simulate elastic solids, there are two advantages in terms of computational efficiency: (1) the TL-SPH equation allows the establishment of a particle linked-list in the initial configuration, saving the time for particle searching; (2) the shell model allows the use of a one-layer particle discrete shell structure, significantly reducing the number of discrete particles. Hence, this study adopts the TL-SPH formulation and the Mindlin–Reissner theory to establish the solid model for the simulation of shell structures. The established TL-SPH shell model will be combined with the weakly compressible SPH fluid model.

4.1. TL-SPH shell model

[Fig. 6](#) shows the SPH model for a single droplet impact on an elastic plate. The initial droplet is a sphere with the diameter D_i . The droplet is modeled by WC-SPH and discretized into WC-SPH particles in a regular and uniform distribution. The elastic plate is modeled by TL-SPH and discretized into one-layer SPH shell particles. The reference configuration of the SPH shell model is defined on the xy plane. The WC-SPH droplet and TL-SPH shell models interact through a contact algorithm, which will be discussed in Section 4.2.

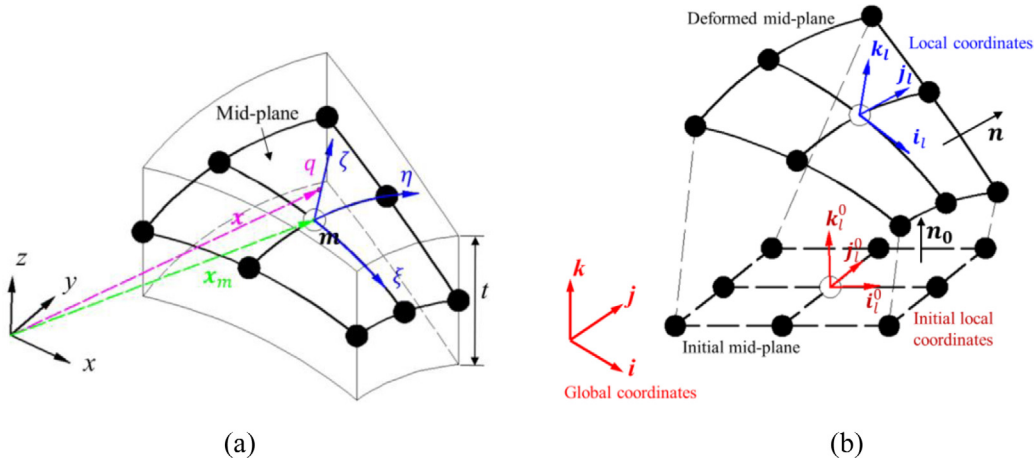


Fig. 7. Discretization of the mid-plane of shell using one layer of SPH particles. (a) Black particles at the mid-plane. (b) Coordinates configuration.

According to the Mindlin–Reissner theory (Hughes and Liu, 1981a,b), the deformation of a shell can be described by a layer of particles distributed on the middle plane. Each particle is assigned a thickness t and has five degrees of freedom (DOF), including three translational and two rotational DOFs for evaluating the bending effect.

As shown in Fig. 7(a), the position vector of a material point q can be expressed as:

$$\mathbf{x}(\xi, \eta, \zeta) = \mathbf{x}_m(\xi, \eta) + \zeta \cdot \mathbf{n}, \quad \zeta \in [-t/2, t/2] \quad (65)$$

where the index m represents the mid-plane, $\mathbf{x}_m$ is the position vector of a mid-plane point, and ζ is the distance from the mid-plane. $\mathbf{n} = (n_x, n_y, n_z)$ is the pseudo-normal vector, which represents the orientation of the mid-plane. In the initial configuration, the position vector of the point q is written as:

$$\mathbf{x}^0(\xi, \eta, \zeta) = \mathbf{x}_m^0(\xi, \eta) + \zeta \cdot \mathbf{n}^0, \quad \zeta \in [-t/2, t/2] \quad (66)$$

Based on Eqs. (65) and (66), the displacement vector $\mathbf{u} = (u, v, w)$ of the point q is calculated by:

$$\mathbf{u}(\xi, \eta, \zeta) = \mathbf{x}(\xi, \eta, \zeta) - \mathbf{x}^0(\xi, \eta, \zeta) = \mathbf{u}_m + \zeta \cdot (\mathbf{n} - \mathbf{n}^0) = \mathbf{u}_m + \zeta \cdot \Delta \mathbf{n}, \quad (67)$$

where $\mathbf{n}^0$ is the pseudo-normal vector in the initial configuration. $\mathbf{u}_m = (u_m, v_m, w_m)$ is the displacement vector of the point at the mid-plane. $\mathbf{n}^0$ is parallel to the global basis $\mathbf{k}$. $\Delta \mathbf{n} = (\Delta n_x, \Delta n_y, \Delta n_z) = \mathbf{n} - \mathbf{n}^0$. The Mindlin–Reissner theory is a thick plate theory, the normal $\mathbf{n}$ is not necessarily perpendicular to the deformed mid-plane. $\mathbf{n}$ is calculated from the evolution history of the rotational degrees of freedom (Maurel and Combescure, 2008).

The Green strain is calculated first in the global coordinate system, then the Euler–Almansi strain can be obtained through the transformation from global to local system. Green strain $\mathbf{E}$ is written as:

$$\mathbf{E} = \frac{1}{2} (\mathbf{F}^T \mathbf{F} + \mathbf{I}) \quad (68)$$

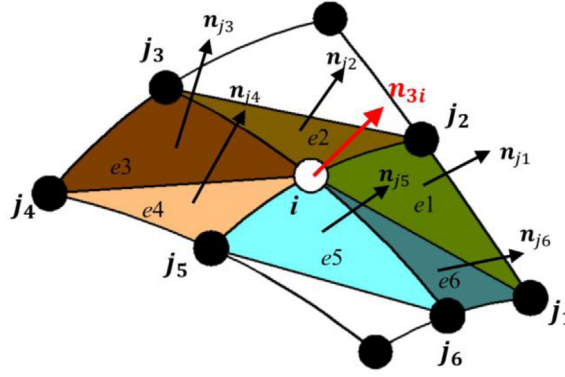
where superscript T represents the transpose of the matrix. $\mathbf{I}$ denotes identity matrix.

Fig. 7(b) defines two coordinate systems, including the fixed global coordinates defined by the identity matrix $\mathbf{I}_g = [\mathbf{i}, \mathbf{j}, \mathbf{k}]$, and the local coordinates defined by $\mathbf{I}_l = [\mathbf{i}_l, \mathbf{j}_l, \mathbf{k}_l]$. A transformation matrix $\mathbf{Q} = [\mathbf{i}_l, \mathbf{j}_l, \mathbf{k}_l]$ is defined to transform the field variable from the global to the local coordinates. $\mathbf{Q}_0 = [\mathbf{i}_l^0, \mathbf{j}_l^0, \mathbf{k}_l^0]$ is the transformation matrix corresponding to the initial configuration. $\mathbf{Q}$ is expressed as (Lin et al., 2014):

$$\mathbf{Q} = \begin{bmatrix} \mathbf{n}_3 \cdot \mathbf{k} + \frac{n_{3y}^2}{1 + \mathbf{n}_3 \cdot \mathbf{k}} & -\frac{n_{3x}n_{3y}}{1 + \mathbf{n}_3 \cdot \mathbf{k}} & n_{3x} \\ -\frac{n_{3x}n_{3y}}{1 + \mathbf{n}_3 \cdot \mathbf{k}} & \mathbf{n}_3 \cdot \mathbf{k} + \frac{n_{3x}^2}{1 + \mathbf{n}_3 \cdot \mathbf{k}} & n_{3y} \\ -n_{3x} & -n_{3y} & n_{3z} \end{bmatrix}, \quad (69)$$

where $\mathbf{k}$ is the unit vector of the z -axis of the global coordinate system. $\mathbf{n}_3 = (n_{3x}, n_{3y}, n_{3z})$ is the normal vector of the mid-plane.

As mentioned above, the shell structure is discretized by a single layer of particles at the mid-plane. For a shell particle i , the transformation matrix $\mathbf{Q}_i$ and normal vector $\mathbf{n}_{3i}$ need to be calculated. $\mathbf{n}_{3i}$ is calculated based on the adjacent particles.

Fig. 8. Calculation of normal vector $\mathbf{n}_{3i}$.

As shown in Fig. 8, the particle i and any two of its adjacent particles form a triangular element. For an inner particle, the number of elements is $n_{elm} = 6$. $\mathbf{n}_{3i}$ can be calculated by area-weighted average of the normal vectors of the elements, as follows:

$$\mathbf{n}_{3i} = \frac{\sum_{a=1}^{n_{elm}} A_{ja} \mathbf{n}_{ja}}{\sum_{j=a}^{n_{elm}} A_{ja}}, \quad \mathbf{n}_{ja} = \mathbf{x}_{ja} \times \mathbf{x}_{j(a-1)}, \quad (70)$$

where $\mathbf{n}_{ja}$ is the normal vector of the element a , $\mathbf{x}_{ja} = \mathbf{x}_{ja} - \mathbf{x}_i$, $\mathbf{x}_{j(a-1)} = \mathbf{x}_{j(a-1)} - \mathbf{x}_i$. After calculating the normal vector $\mathbf{n}_{3i}$, the transformation matrix $\mathbf{Q}_i$ can be obtained using Eq. (69).

Considering an initial point at the shell mid-plane $\mathbf{x}_m^0 = (x_m^0, y_m^0, z_m^0)$, it moves to $\mathbf{x}_m = (x_m, y_m, z_m)$ in the current configuration, the deformation gradient tensor $\mathbf{F}$ can be defined as (Ming et al., 2013):

$$\mathbf{F} = \frac{\partial \mathbf{x}_m}{\partial \mathbf{x}_m^0} = \begin{bmatrix} \frac{\partial x_m}{\partial x_m^0} & \frac{\partial x_m}{\partial y_m^0} & \frac{\partial x_m}{\partial z_m^0} \\ \frac{\partial y_m}{\partial x_m^0} & \frac{\partial y_m}{\partial y_m^0} & \frac{\partial y_m}{\partial z_m^0} \\ \frac{\partial z_m}{\partial x_m^0} & \frac{\partial z_m}{\partial y_m^0} & \frac{\partial z_m}{\partial z_m^0} \end{bmatrix}. \quad (71)$$

The deformation gradient tensor is also written as:

$$\mathbf{F} = \frac{\partial (\mathbf{x}_m - \mathbf{x}_m^0)}{\partial \mathbf{x}_m^0} + \frac{\partial \mathbf{x}_m^0}{\partial \mathbf{x}_m^0} = \frac{\partial \mathbf{u}_m}{\partial \mathbf{x}_m^0} + \mathbf{I} = \mathbf{L} + \mathbf{I}. \quad (72)$$

Hence, the Green strain tensor can be written as:

$$\mathbf{E} = \frac{1}{2} (\mathbf{L}^T \mathbf{L} + \mathbf{L} + \mathbf{L}^T) \quad (73)$$

where $\mathbf{L}$ is displacement gradient tensor, $\mathbf{L} = \frac{\partial \mathbf{u}_m}{\partial \mathbf{x}_m^0}$, $\mathbf{u}_m = (u_m, v_m, w_m)$.

For a shell structure, the Green strain can be divided into three parts: constant membrane strain $\mathbf{E}_m$, constant shear strain $\mathbf{E}_s$, and bending strain $\zeta \mathbf{E}_b$ that varies linearly along the thickness direction.

The elastic substrate (i.e., cantilever) considered in this study may experience large deflection due to droplet impact. Hence, the nonlinear shearing strain and bending strain are not considered in the present SPH shell model (Maurel and Combescure, 2008; Ming et al., 2013). Each part of Green strain is written as:

$$\mathbf{E}_m + \mathbf{E}_s = \begin{bmatrix} \frac{\partial u_m}{\partial x_0} & \frac{1}{2} \left(\frac{\partial u_m}{\partial y_0} + \frac{\partial v_m}{\partial x_0} \right) & \frac{1}{2} \left(\Delta n_x + \frac{\partial w_m}{\partial x_0} \right) \\ \dots & \frac{\partial v_m}{\partial y_0} & \frac{1}{2} \left(\Delta n_y + \frac{\partial w_m}{\partial y_0} \right) \\ \dots & \dots & 0 \end{bmatrix}, \quad (74)$$

$$\text{and } \mathbf{E}_b = \begin{bmatrix} \frac{\partial (\Delta n_x)}{\partial x_0} & \frac{1}{2} \left(\frac{\partial (\Delta n_x)}{\partial y_0} + \frac{\partial (\Delta n_y)}{\partial x_0} \right) & 0 \\ \dots & \frac{\partial (\Delta n_y)}{\partial y_0} & 0 \\ \dots & \dots & 0 \end{bmatrix} \quad (75)$$

where $\Delta \mathbf{n} = (\Delta n_x, \Delta n_y, \Delta n_z)$.

Correspondingly, Euler-Almansi strain $\boldsymbol{\varepsilon}$, which includes membrane strain $\boldsymbol{\varepsilon}_m$, bending strain $\boldsymbol{\varepsilon}_b$ and shear strain $\boldsymbol{\varepsilon}_s$, can be obtained through following transformations:

$$\begin{cases} \boldsymbol{\varepsilon}_m = \mathbf{Q} \mathbf{F}^{-T} \mathbf{E}_m \mathbf{F}^{-1} \mathbf{Q}^T \\ \boldsymbol{\varepsilon}_s = \mathbf{Q} \mathbf{F}^{-T} \mathbf{E}_s \mathbf{F}^{-1} \mathbf{Q}^T \\ \boldsymbol{\varepsilon}_b = \mathbf{Q} \mathbf{F}^{-T} \mathbf{E}_b \mathbf{F}^{-1} \mathbf{Q}^T \end{cases} \quad (76)$$

The isotropic elastic constitutive model is adopted. Considering the plane-stress assumption, the membrane stress $\boldsymbol{\sigma}_m$, bending stress $\boldsymbol{\sigma}_b$, and transverse shear stress $\boldsymbol{\sigma}_s$ are written as:

$$\boldsymbol{\sigma}_m = \mathbf{H} \cdot \boldsymbol{\varepsilon}_m, \quad \boldsymbol{\sigma}_b = \zeta \mathbf{H} \cdot \boldsymbol{\varepsilon}_b, \quad \boldsymbol{\sigma}_s = k \mathbf{H}_s \cdot \boldsymbol{\varepsilon}_s, \quad (77)$$

where $\mathbf{H} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$, $\mathbf{H}_s = G \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\boldsymbol{\varepsilon}_m = \begin{pmatrix} \varepsilon_{mxx} \\ \varepsilon_{myy} \\ 2\varepsilon_{mxy} \end{pmatrix}$, $\boldsymbol{\varepsilon}_b = \begin{pmatrix} \varepsilon_{bxx} \\ \varepsilon_{byy} \\ 2\varepsilon_{bxy} \end{pmatrix}$, $\boldsymbol{\varepsilon}_s = \begin{pmatrix} 2\varepsilon_{sxz} \\ 2\varepsilon_{syx} \end{pmatrix}$, E is Young's modulus,

ν is Poisson's ratio. G is shear modulus, $G = \frac{E}{2(1+\nu)}$. k is shear correction factor, for a rectangular section, $k = 5/6$.

To construct the equilibrium equations of a shell, the resultant generalized force should be calculated by integrating each stress component along the thickness direction. The resultant generalized forces including membrane force $\mathbf{N}$, bending moment $\mathbf{M}$ and shear force $\mathbf{T}$, are obtained by following integrations:

$$\begin{cases} N_{ij} = \int_{-t/2}^{t/2} \sigma_{mij} d\zeta \\ M_{ij} = \int_{-t/2}^{t/2} \sigma_{bij} \zeta d\zeta \\ T_{ij} = \int_{-t/2}^{t/2} \sigma_{sij} d\zeta \end{cases} \quad (78)$$

Based on Eqs. (78) and (77), the resultant generalized forces are written as:

$$\begin{pmatrix} N_{xx} \\ N_{yy} \\ N_{xy} \end{pmatrix} = \begin{pmatrix} \frac{E}{1-\nu^2} t & \frac{E\nu}{1-\nu^2} t & 0 \\ \frac{E\nu}{1-\nu^2} t & \frac{E}{1-\nu^2} t & 0 \\ 0 & 0 & \frac{E}{1+\nu} \frac{t}{2} \end{pmatrix} \begin{pmatrix} \varepsilon_{mxx} \\ \varepsilon_{myy} \\ 2\varepsilon_{mxy} \end{pmatrix}, \quad (79)$$

$$\begin{pmatrix} M_{xx} \\ M_{yy} \\ M_{xy} \end{pmatrix} = \begin{pmatrix} \frac{E}{1-\nu^2} \frac{t^3}{12} & \frac{E\nu}{1-\nu^2} \frac{t^3}{12} & 0 \\ \frac{E\nu}{1-\nu^2} \frac{t^3}{12} & \frac{E}{1-\nu^2} \frac{t^3}{12} & 0 \\ 0 & 0 & \frac{E}{1+\nu} \frac{t^3}{24} \end{pmatrix} \begin{pmatrix} \varepsilon_{bxx} \\ \varepsilon_{byy} \\ 2\varepsilon_{bxy} \end{pmatrix}, \quad (80)$$

$$\begin{pmatrix} T_{xz} \\ T_{yz} \end{pmatrix} = \begin{pmatrix} \frac{kE}{2(1+\nu)} t & 0 \\ 0 & \frac{kE}{2(1+\nu)} t \end{pmatrix} \begin{pmatrix} 2\varepsilon_{sxz} \\ 2\varepsilon_{syx} \end{pmatrix}. \quad (81)$$

Then, in the local coordinate system, the differential equilibrium equation of the shell is written as:

$$\begin{cases} \rho_m \ddot{u} = N_{xx,x} + N_{xy,y} \\ \rho_m \ddot{v} = N_{yx,x} + N_{yy,y} \\ \rho_m \ddot{w} = T_{xz,x} + T_{yz,y} \\ I_b \ddot{\varphi} = M_{xx,x} + M_{xy,y} - T_{xz} \\ I_b \ddot{\theta} = M_{yx,x} + M_{yy,y} - T_{yz} \end{cases}, \quad (82)$$

where comma represents partial differentiation, for example, $N_{xx,x} = \frac{\partial N_{xx}}{\partial x}$, $\rho_m = \int_{-t/2}^{t/2} \rho d\zeta = t\rho$, $I_b = \int_{-t/2}^{t/2} t\rho d\zeta = \frac{t^3}{12}\rho$. Eq. (82) can be written as a matrix form:

$$\begin{cases} \rho_m \ddot{\mathbf{u}}_l = \text{div}_l (\mathbb{N}_l) + \mathbf{b}_l \\ I_b \ddot{\boldsymbol{\theta}}_l = \text{div}_l (\mathbb{M}_l) + \mathbf{T}_l \end{cases} \quad (83)$$

where $\mathbb{N}_l = \begin{bmatrix} N_{xx} & N_{xy} & T_{xz} \\ N_{yx} & N_{yy} & T_{yz} \\ T_{zx} & T_{zy} & 0 \end{bmatrix}$, $\mathbb{M}_l = \begin{bmatrix} M_{xx} & M_{xy} & 0 \\ M_{yx} & M_{yy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\mathbf{T}_l = \begin{bmatrix} -T_{xz} \\ -T_{yz} \\ 0 \end{bmatrix}$. The subscript l denotes the local coordinate system. In Eq. (83), the external force vector $\mathbf{b}_l$ is considered.

To establish the discrete forms of the equilibrium equations under the TL-SPH framework, the Cauchy stress should be transformed from the local to global system through $\mathbf{Q}^T \mathbb{N}_l \mathbf{Q}$ and $\mathbf{Q}^T \mathbb{M}_l \mathbf{Q}$. From local to global system, the equilibrium equations Eq. (83) are converted to:

$$\begin{cases} \rho_m \ddot{\mathbf{U}} = \text{DIV} (\mathbb{N}_0) + \mathbf{b}_0 \\ I_b \ddot{\boldsymbol{\Theta}} = \text{DIV} (\mathbb{M}_0) + \mathbf{T}_0 \end{cases}, \quad (84)$$

where $\mathbb{N}_0 = \mathbf{J} \mathbf{F}^{-1} \mathbf{Q}^T \mathbb{N}_l \mathbf{Q}$, $\mathbb{M}_0 = \mathbf{J} \mathbf{F}^{-1} \mathbf{Q}^T \mathbb{M}_l \mathbf{Q}$, $\mathbf{T}_0 = \mathbf{J} \mathbf{Q}^T \mathbf{T}_l$, $\mathbf{b}_0 = \mathbf{J} \mathbf{Q}^T \mathbf{b}_l$. $\mathbf{J}$ is determinant of the deformation gradient, i.e., $\mathbf{J} = |\mathbf{F}|$.

The total Lagrangian SPH formulation takes the initial configuration as the reference space, where the kernel function $W_0(|\mathbf{x}_{ij}^0|, h_{ij})$ and $\nabla_{\mathbf{x}_i^0} W(\mathbf{x}_i^0 - \mathbf{x}_j^0, h_{ij})$ are calculated. For each SPH shell particle, the deformation gradient tensor $\mathbf{F}_i$ is calculated as:

$$\mathbf{F}_i = \begin{bmatrix} \sum_{j=1}^{N_i^s} (x_{mj} - x_{mi}) \frac{\partial W_{ij}^0}{\partial x_{0i}} A_{0j} & \sum_{j=1}^{N_i^s} (x_{mj} - x_{mi}) \frac{\partial W_{ij}^0}{\partial y_{0i}} A_{0j} & n_x \\ \sum_{j=1}^{N_i^s} (y_{mj} - y_{mi}) \frac{\partial W_{ij}^0}{\partial x} A_{0j} & \sum_{j=1}^{N_i^s} (y_{mj} - y_{mi}) \frac{\partial W_{ij}^0}{\partial y_{0i}} A_{0j} & n_y \\ \sum_{j=1}^{N_i^s} (z_{mj} - z_{mi}) \frac{\partial W_{ij}^0}{\partial x_{0i}} A_{0j} & \sum_{j=1}^{N_i^s} (z_{mj} - z_{mi}) \frac{\partial W_{ij}^0}{\partial y_{0i}} A_{0j} & n_z \end{bmatrix}, \quad (85)$$

where $\nabla_{\mathbf{x}_i^0} W(\mathbf{x}_i^0 - \mathbf{x}_j^0, h_{ij}) = \left(\frac{\partial W_{ij}^0}{\partial x_{0i}}, \frac{\partial W_{ij}^0}{\partial y_{0i}} \right)$. $A_{0j} = d_{ini}^2$, is the area of the SPH shell particle.

Similarly, for each SPH shell particle, the displacement gradient tensor is calculated as:

$$\mathbf{L}_i = \begin{bmatrix} \sum_{j=1}^{N_i^s} (u_{mj} - u_{mi}) \frac{\partial W_{ij}^0}{\partial x_{0i}} A_{0j} & \sum_{j=1}^{N_i^s} (u_{mj} - u_{mi}) \frac{\partial W_{ij}^0}{\partial y_{0i}} A_{0j} & \Delta n_x \\ \sum_{j=1}^{N_i^s} (v_{mj} - v_{mi}) \frac{\partial W_{ij}^0}{\partial x} A_{0j} & \sum_{j=1}^{N_i^s} (v_{mj} - v_{mi}) \frac{\partial W_{ij}^0}{\partial y_{0i}} A_{0j} & \Delta n_y \\ \sum_{j=1}^{N_i^s} (w_{mj} - w_{mi}) \frac{\partial W_{ij}^0}{\partial x_{0i}} A_{0j} & \sum_{j=1}^{N_i^s} (w_{mj} - w_{mi}) \frac{\partial W_{ij}^0}{\partial y_{0i}} A_{0j} & \Delta n_z \end{bmatrix}, \quad (86)$$

where $\mathbf{u}_m = (u_m, v_m, w_m)$, $\mathbf{n} = (\Delta n_x, \Delta n_y, \Delta n_z)$.

The Green strain can be calculated based on Eqs. (74) and (75). The SPH discrete equation for simulating the motion of the shell structure is written as:

$$\begin{cases} \ddot{\mathbf{u}}_i = \sum_{j=1}^{N_i^s} \left(\frac{\mathbb{N}_{0i}}{\rho_{mi}^2} + \frac{\mathbb{N}_{0j}}{\rho_{mj}^2} \right) \nabla_{\mathbf{x}_i^0} W(\mathbf{x}_i^0 - \mathbf{x}_j^0, h_{ij}) m_j + \frac{\mathbf{b}_{0i}}{\rho_{mi}} \\ \ddot{\Theta}_i = \sum_{j=1}^{N_i^s} \left(\frac{\mathbb{M}_{0i}}{I_{bi}^2} + \frac{\mathbb{M}_{0j}}{I_{bj}^2} \right) \nabla_{\mathbf{x}_i^0} W(\mathbf{x}_i^0 - \mathbf{x}_j^0, h_{ij}) m_j + \frac{\mathbf{T}_{0i}}{I_{bi}} \end{cases}, \quad (87)$$

where N_i^s represents the total number of particles in the support domain of particle i . m_j is mass of the shell particle, $m_j = \rho_{mj} A_{0j} = \rho_s t d_{ini}^2$.

To make the calculation stable, the artificial viscous force is also used in the SPH shell model. The viscous force between any particle pair ij is decomposed into two parts, namely the in-plane force S_{vij} and the transverse shear component T_{vij} . They can be written as (Lin, 2014):

$$S_{vij} = \frac{-\alpha_{\Pi} \bar{c}_{ij} \phi_{ij} + \beta_{\Pi} \phi_{ij}^2}{\bar{\rho}_{mij}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \phi_{ij} < 0, \quad (88)$$

$$T_{vij} = \frac{-\gamma_{\Pi} \bar{c}_{ij} \psi_{ij}}{\bar{\rho}_{mij}} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \psi_{ij} < 0, \quad (89)$$

where $\phi_{ij} = \frac{h_{ij}[(v_{xi}-v_{xj})(x_i-x_j)+(v_{yi}-v_{yj})(y_i-y_j)]}{|\mathbf{x}_{ij}|^2+0.01h_{ij}^2}$, $\psi_{ij} = \frac{h_{ij}(v_{zi}-v_{zj})(z_i-z_j)}{|\mathbf{x}_{ij}|^2+0.01h_{ij}^2}$.

Further, the artificial viscous force of particle pair in global coordinates is written as:

$$\mathbf{N}_{vij} = J_i \mathbf{Q}_i^T (S_{vij} + T_{vij}) \mathbf{Q}_i \mathbf{G}_i^T \quad (90)$$

where $J_i = |\mathbf{F}_i|$ the determinant of the deformation gradient tensor.

The SPH discrete form of the linear equilibrium equation is finally written as:

$$\ddot{\mathbf{u}}_i = \sum_{j=1}^{N_i^s} \left(\frac{\mathbb{N}_{0i}}{\rho_{mi}^2} + \frac{\mathbb{N}_{0j}}{\rho_{mj}^2} - N_{vij} \right) \nabla_{\mathbf{x}_i^0} W(\mathbf{x}_i^0 - \mathbf{x}_j^0, h_{ij}) m_j + \frac{\mathbf{b}_{0i}}{\rho_{mi}} \quad (91)$$

4.2. fluid-shell interaction model

The interaction between the droplet and the substrate is modeled by the contact algorithm. A particle-surface contact force model is established. If a fluid SPH particle is located within a certain distance (d_0) of the shell surface, it is identified as a contact particle. To implement contact algorithm, the first step is to determine whether the particle is in contact. As shown in Fig. 9, since the shell surface is located $t/2$ above the mid-plane, the real shell surface is constructed according to the SPH shell particles at the mid-plane. The projection point o of fluid particle i on the shell surface is found, and the vertical distance d_p to shell surface is calculated. If $d_p < d_0$, the particle i is in contact with the shell surface, and the contact force is calculated by the following equation:

$$\mathbf{F}_i = (1 - \chi) \left[\frac{2|d_0 - d_p|}{(\Delta t)^2} \right] m_i \mathbf{n}_i, \quad (92)$$

where Δt is time step, m_i is mass of particle i . d_0 is distance threshold for contact, $d_0 = d_{ini}$. $\mathbf{n}_i$ is the unit normal vector of the shell surface, pointing outside the surface.

According to the principle of action and reaction, the contact force of the shell subjected to the fluid particle i is $-\mathbf{F}_i$. This force should be dispersed to the adjacent shell particles j_1 , j_2 and j_3 , namely $\mathbf{F}_{j1}$, $\mathbf{F}_{j2}$ and $\mathbf{F}_{j3}$. According to the force and moment balance, the contact force on the shell particles is calculated as follows:

$$\begin{cases} \frac{\mathbf{F}_{j1}}{|\mathbf{F}_{j1}|} = \frac{\mathbf{F}_{j2}}{|\mathbf{F}_{j2}|} = \frac{\mathbf{F}_{j3}}{|\mathbf{F}_{j3}|} = -\mathbf{n}_i \\ |\mathbf{F}_{j1}| + |\mathbf{F}_{j2}| + |\mathbf{F}_{j3}| = |\mathbf{F}_i| \\ |\mathbf{F}_{j1}| \mathbf{n}_i \times \mathbf{S}_1 + |\mathbf{F}_{j2}| \mathbf{n}_i \times \mathbf{S}_2 + |\mathbf{F}_{j3}| \mathbf{n}_i \times \mathbf{S}_3 = 0 \end{cases} \quad (93)$$

where $\mathbf{S}_1$, $\mathbf{S}_2$ and $\mathbf{S}_3$ are the relative position vectors from the point o' to the shell particles j_1 , j_2 and j_3 .

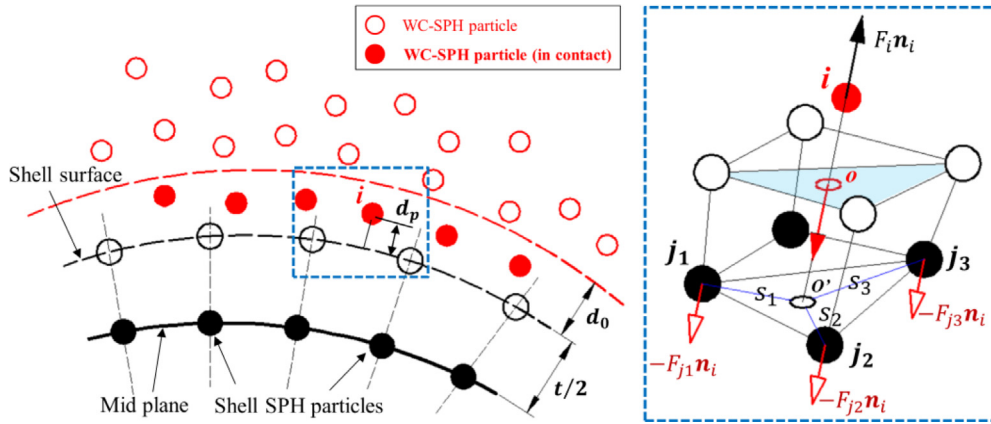


Fig. 9. The contact algorithm between the droplet and the SPH shell.

4.3. Time integration scheme

The explicit time integration scheme is used to solve the 3D fluid–solid coupling SPH model. The WC-SPH and TL-SPH shell models are solved in the same time integration loop but use different schemes to update variables. The displacement of a shell particle at the $n + 1$ th step is obtained from the variables at the n th and $n-1$ th steps, and is calculated by the following equation (Lin, 2014):

$$\mathbf{U}^{n+1} = \frac{4}{2 + \alpha \Delta t} \mathbf{U}^n + \frac{\alpha \Delta t - 2}{2 + \alpha \Delta t} \mathbf{U}^{n-1} + \frac{2 \Delta t^2}{2 + \alpha \Delta t} \ddot{\mathbf{U}}^n. \quad (94)$$

where the current acceleration $\ddot{\mathbf{U}}^n$ is calculated by Eq. (91) based on field variables at the n th. The prediction–correction scheme is adopted to update variables of WC-SPH model.

5. Results and discussion

5.1. Validation of surface tension model

5.1.1. Droplet deformation process

In this section, an example of square droplet evolution is used to verify the correctness of the surface force model. The side length of the square droplet is set to 1.0 mm, the physical parameters are set according to the water. The density is set to 1000 kg/m^3 , and the surface tension coefficient is set to 0.072 N/m . Three kinds of particle spacing are adopted, the number of side particles is 25, 40, 80 particles respectively, the corresponding spacing is 0.04, 0.025, 0.0125 mm, and the number of particles is 625, 1600, 6400. After the calculation begins, the droplet experiences a period of oscillation, and the kinetic energy dissipates to the minimum under the action of viscous force, etc., and the droplet finally reaches a quasi-static circle, as shown in Fig. 10.

Fig. 11 shows the curvature distribution of boundary particles at different times. At 0.03 ms, although the sharp corner of the square droplet has been smoothed under the action of surface force, the corner point still has a high curvature value. After reaching the quasi-static state at 4.2 ms, the droplet morphology is nearly round, and the curvature of surface particles is close to the theoretical value $1/R$ ($=1772.5$). After several contractive oscillations, the droplet evolves into a quasi-static state. When the droplet reaches this state, the surface energy of the droplet decreases to its lowest level, and the curvature of the surface particles is close to the theoretical value $1/R$. According to Laplace's law, the theoretical solution of the average pressure inside the droplet is 132.9 Pa. Fig. 12 shows the change curve of the internal mean pressure with time during the evolution of the square droplet. As can be seen from the figure, with the evolution of droplet morphology, the pressure also approaches a stable value and finally fluctuates slightly around 137 Pa, reaching a quasi-static process, which is in good agreement with the theoretical solution.

Deformation of a cubic liquid droplet is simulated by the WC-SPH model in 3D. The side length of the initial droplet is 6.25×10^{-3} and 125,000 particles are used in the simulation. The coefficient of surface tension is set to 2.5. The viscosity of the liquid is 1.0×10^{-6} . Fig. 13 shows the snapshot of oscillation of cubic liquid droplet at $t = 0.0005, 0.001, 0.0015, 0.002, 0.003$ and 0.004 . Due to surface tension force on the surface of cubic liquid droplet, the eight corners of cubic liquid droplet deform first. The liquid drop evolves into dumbbell shape and has larger velocity at the corners at $t = 0.0005$. The eight round corners of liquid drop continue to contract and have a shape of non-equilibrium sphere at $t = 0.001$. The liquid drop becomes a drop with a shape of the diamond at $t = 0.0015$. Due to the viscous dissipation, the liquid drop evolves into a sphere at equilibrium at $t = 0.004$. During the oscillation, cubic liquid drop evolves into dumbbell, non-equilibrium sphere, diamond, sphere at equilibrium

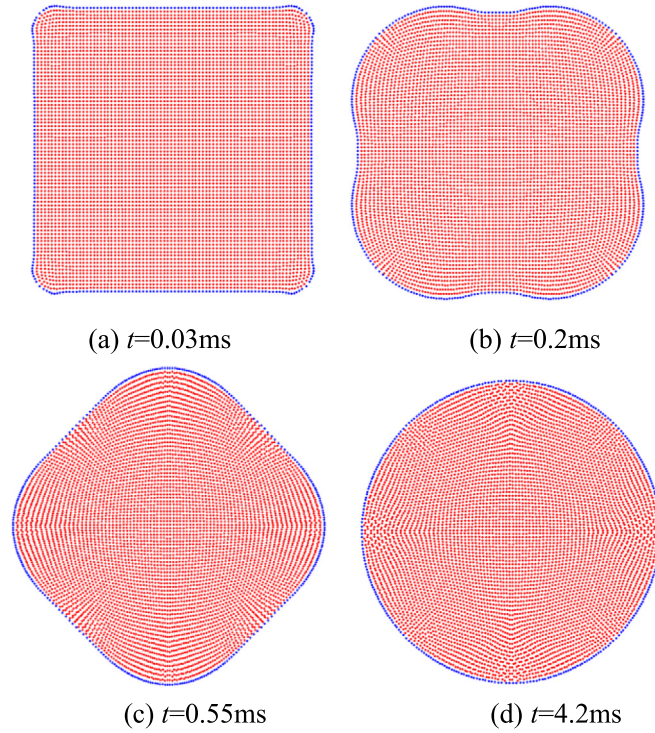


Fig. 10. Square droplet deformation under the action of surface tension. Blue represents the boundary particles and red represents the inner particles. . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

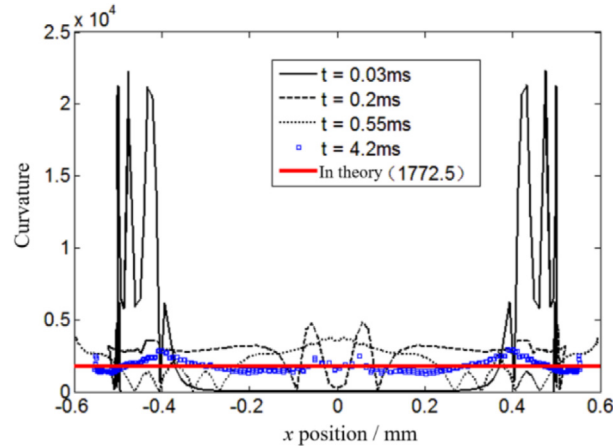


Fig. 11. Distribution of curvature of boundary particles during the deformation process.

5.1.2. Droplet impact on rigid and smooth surfaces

The SPH model is used to simulate the dynamic process of liquid droplets impact on a rigid substrate. To compare the simulation results with available experimental data, this section focuses on the super-hydrophobic surface. Contact time is an important factor to evaluate the super-hydrophobic property of the surface, and it is defined as the duration from contact to bouncing. By balancing the inertial force and capillary force of the droplet, the contact time τ_c can be calculated as Richard et al. (2002), $\tau_c = \sqrt{\rho D_i^3 / \gamma}$, where D_i is the droplet diameter, and γ is the surface tension. It shows that the contact time is related to the droplet diameter and surface tension, but is independent from the impact velocity.

Four test cases are configured and simulated. Simulation parameters are given in Table 1. By setting the droplet diameter and impact velocity, different We are obtained, i.e., $We = 33, 49.5, 66, 76.2$. We is a dimensionless parameter commonly used to analyze droplet impact mechanisms, $We = \frac{\rho V_i^2 D_i}{\gamma}$, denoting the ratio of inertial force and surface tension

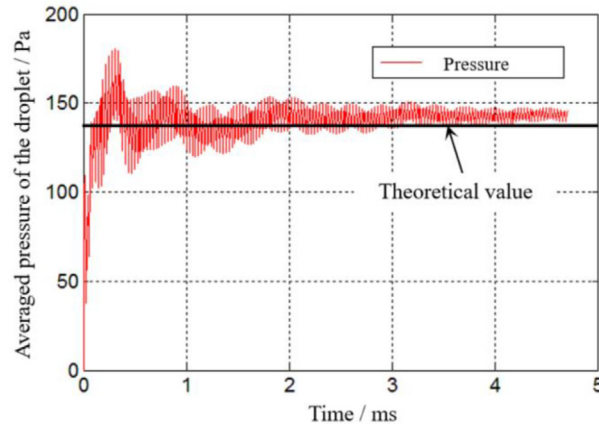


Fig. 12. Time history of pressure during the deformation process.

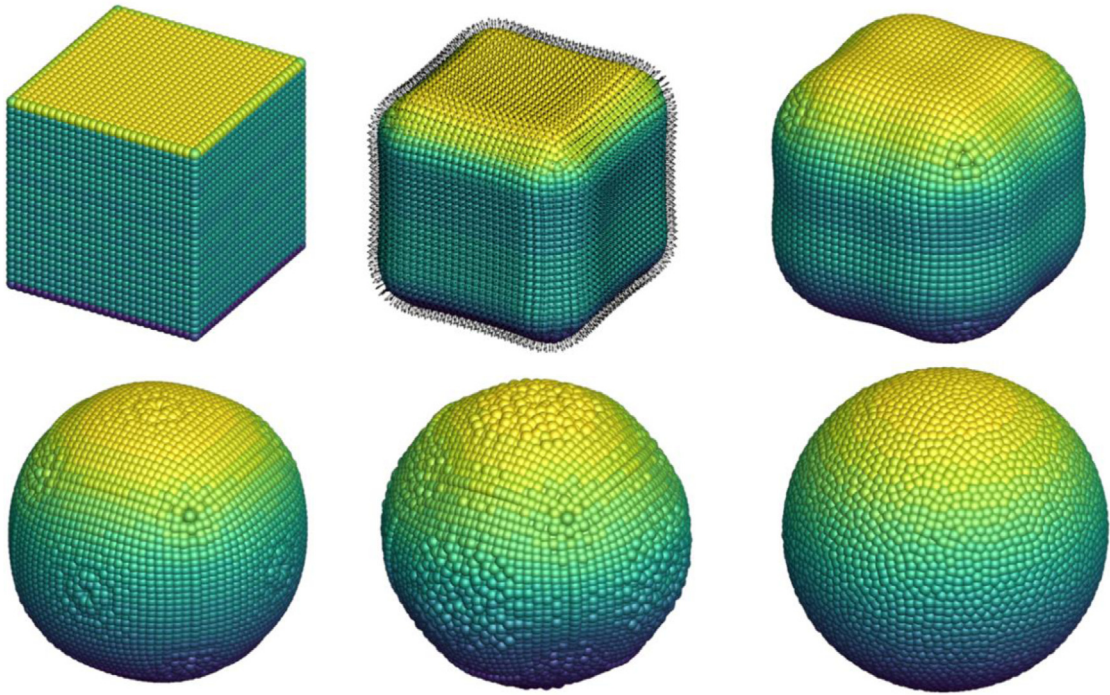


Fig. 13. 3D Simulation results of cubic droplet deformation process.

force. The density is set to 1000 kg/m^3 , and the kinematic viscosity is set to $1.0 \times 10^{-6} \text{ m}^2/\text{s}$. The diameter of droplet is set to 1.8, 2.4 and 2.8 mm, and the impact velocity is 1.0 and 1.4 m/s. In this section, the surface is assumed to rigid, that is, only WC-SPH model is used. The particle spacing is set to $1/25.0D_i$. According to CFL criterion, the time step Δt is set to $1 \times 10^{-6} \text{ s}$. Because We is small, no severe splashing is observed. The droplet can rebound after impact. The dynamic process of droplet impact is predicted by the WC-SPH model, and the maximum spreading diameter D_{max} , the spreading factor ($\beta = D/D_i$) and contact time are obtained, which are compared with the experimental results to validate the model.

In the range $We = 33 \sim 76.2$, droplets can achieve the complete rebound without breakup, which is also consistent with the experimental observations (Upadhyay et al., 2021). As We increases, the spreading diameter of the droplets also increases. The spreading coefficient $\beta = D_{max}/D_i$ is proportional to We (Clanet et al., 2004), i.e., $\beta = \alpha We^{0.5}$. The SPH simulation and experimental results for spreading coefficient and contact time are given in Table 1 and Fig. 14. It shows that the results obtained from the simulation is in good agreement with the experiment.

Table 1
Simulation parameters and results of droplets impact on a rigid surface.

Case No.	D_i/mm	$V_i/\text{m/s}$	We	D_{max}/mm		β_{max}	τ_c/ms	
				SPH	Exp. Lin (2014)		SPH	Exp. Lin (2014)
1	2.4	1.0	33	4.865	4.92	2.01	12.685	13
2	1.8	1.4	49.5	4.287	4.34	2.38	7.724	7.89
3	2.4	1.4	66	6.229	6.6	2.60	12.693	13
4	2.8	1.4	76.2	7.885	7.97	2.81	15.875	16.28

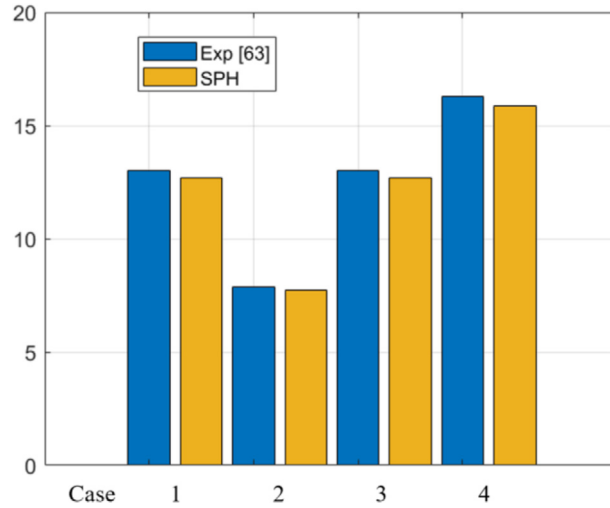


Fig. 14. Comparison of simulation and experimental results of contact time of droplet impact on rigid substrate.

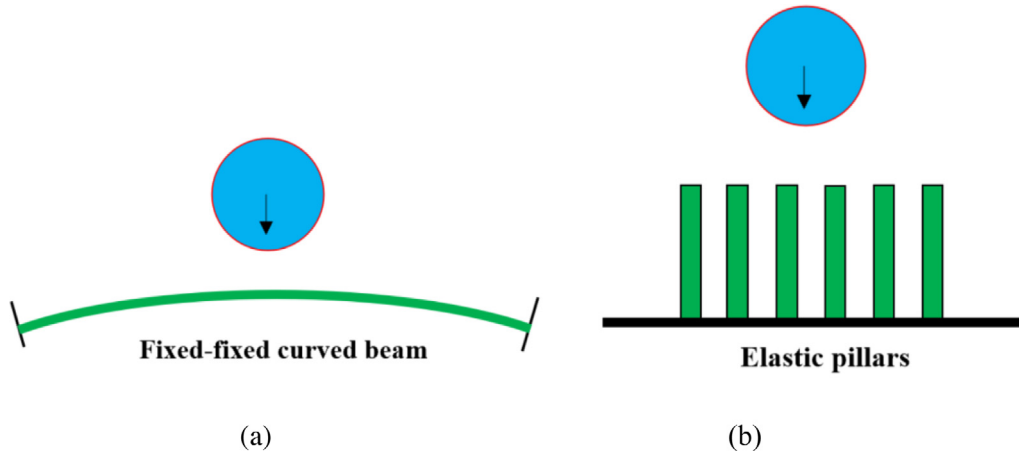


Fig. 15. Schematic diagram of droplet impact on three types of flexible substrates.

5.2. 2D simulation of droplet impact on flexible substrates

In this section, the 2D WC-TL SPH model is used to simulate the process of droplet impact on two types of flexible substrates, as shown in Fig. 15. These are a curved beam and a row of flexible micro-pillars. A droplet impact on these substrates causes elastic deformation, which in turn affects the spreading and bouncing behavior of the droplets. The results demonstrate the applicability of the model to simulating droplet impact on a flexible substrate with complex structures.

5.2.1. Droplet impact on a fixed-fixed and curved beam

The WC-TL SPH model is used to simulate the dynamic process of droplet impact on a curved beam. The radius R of the beam is set to 14.5 mm, the arch height H is set to 4.0 mm, and the arch span W is set to 10.0 mm. The two ends of

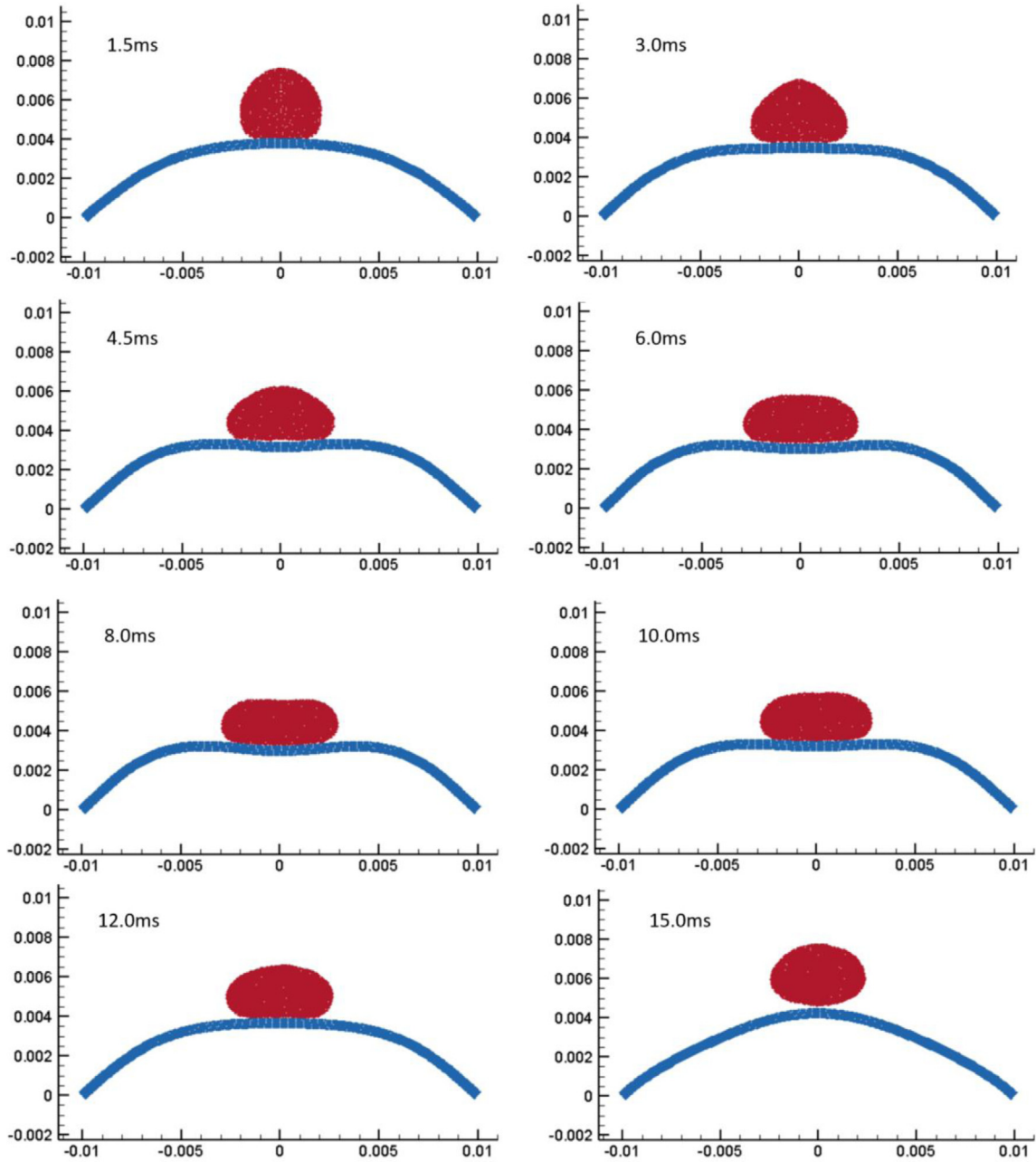


Fig. 16. Simulation results of a droplet impact on a curved beam ($D_i = 4.0$ mm, $V_i = 0.5$ m/s).

the beam are fixed constraints. The droplets impacts on the middle of the beam from top to bottom, causing the beam to deform downward. The elasticity modulus is set to 1.2 MPa, and Poisson's ratio is 0.4, thus the shear modulus and bulk modulus are 0.427 MPa and 2.0 MPa, respectively. At the initial moment, the beam is in a stable position, and the beam deforms under the action of the liquid drops. After the liquid drops bounce back, the beam will return to the position before the deformation under the action of its own elasticity.

The droplet experiences the process of impact, spread, retraction and rebound. How the droplet interacts with the deformation of the beam is difficult to explain through theoretical modeling. The model established in this work is used to simulate the droplet impact on the curved beam under the conditions of impact velocity ranging from 0.3 m/s to 1.0 m/s and droplet diameter of 4.0 mm. Fig. 16 shows the simulation results at different moments when the impact velocity is 0.5 m/s. Fig. 17 shows the change curve of the position of the middle beam with time. The characteristic time of droplet and beam is similar. The droplet reaches its maximum spreading diameter around 8.0 ms, and the beam is also in the

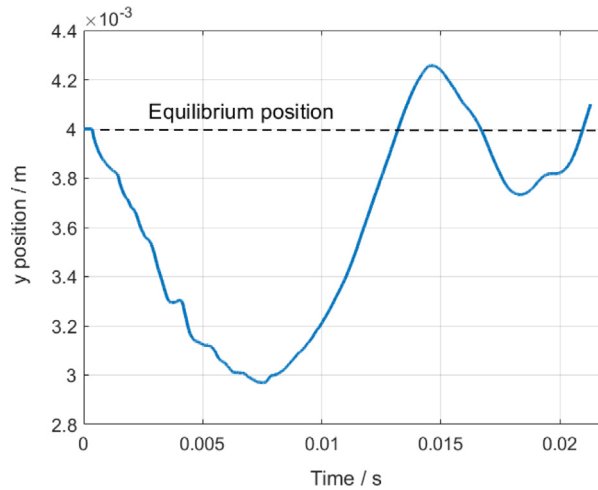


Fig. 17. Time history of the y position of the middle of the beam ($D_i = 4.0$ mm, $V_i = 0.5$ m/s).

position of maximum deformation. The y position coordinate of the beam midpoint is the minimum. After 8 ms, the beam began to bounce back from the lowest position, and the droplet began to retraction. At 13 ms, the droplet returns to its initial position, but continues to move upward due to inertia. At 15 ms, the beam at the highest position was out of contact with the droplet, and the droplet bounced upward. The beam then returns from the highest position to the initial position and oscillates up and down near the initial position for several periods.

Note that when the droplet is out of contact with the beam, the retraction process is not completed, so that the droplet bounce back is not pear-shaped, but close to the pan-cake shape. The maximum displacement of the midpoint of the beam is about 3.0 mm, which is about 3/4 of the droplet diameter. As shown in Fig. 21(a), the upper part of the beam section mainly bears compressive stress and the lower part mainly bears tensile stress. Under the impact of 0.5 m/s, the beam has obvious deformation, but the vibration attenuation rate is very high, which is mainly due to the steady-state structure of the beam. In other words, under droplet impact, the main response of the fixed curved beam at both ends is deformation, while the vibration response is not dominant.

Fig. 18 shows the simulation results at different moments when the impact velocity is 1.0 m/s. Fig. 19 shows the change curve of the position of the middle beam with time. The droplet reaches its maximum spreading diameter at 6.0 ms, while the middle of the beam still moves downward. At 18 ms, the beam reaches its maximum deformation. As shown in Fig. 19, the maximum deformation of the beam exceeds the line $y = 0$. After 18 ms, the beam began to change from the maximum deformation position to the initial position. After 6.0 ms, the droplet with the maximum spreading diameter began to retract. Due to the continuous deformation of the beam, the droplet lacks the upward rebound force during the retraction, causing it to evolve into a circular shape and lose most of its kinetic energy. With the recovery of the beam, the upward motion speed of the beam increases continuously, providing upward motion force for the droplet, allowing the droplet to finally break off the surface of the beam and complete the rebound.

Due to the dynamic and nonlinear characteristics of the contact and motion between the droplet and the beam, the maximum deformation of the beam midpoint is also nonlinear to the impact velocity of the droplet. As shown in Fig. 20, the lowest position of the beam midpoint (corresponding to the maximum deformation of the beam) decreases with the increase of impact velocity and shows a parabolic trend. Fig. 21 shows the stress distribution of the beam under large deformation when the impact velocity is 0.5 and 1.0 m/s respectively. The impact of the droplet on the midpoint of the beam makes the middle of the beam deform downward, showing a mode of two peaks and one trough, which is unstable. Even if the midpoint of the beam can pass through the $y = 0$ line downward, the deformed beam can still return to the initial and stable state.

5.2.2. Droplet impact on a row of elastic pillars

This section is concerned with the process of droplet impingement on a micro-pillar substrate. The micro-pillar substrate is a typical micro-structure that is commonly used to regulate the hydrophobic properties of the surface. It was shown that droplets impacting the micro-pillar substrate undergo pancake-like rebound, which can significantly reduce the contact time of droplets. Studying the interaction process between droplets and micro-pillars is the key to understand this physical mechanism. A row of uniformly arranged micro-pillars with a width w of 0.5 mm and a length l of 3.0 mm and 6.0 mm is used in the simulation. The elasticity modulus of the micro-pillar is set to 1.2 MPa, and Poisson's ratio is set to 0.4, thus the shear modulus and bulk modulus are 0.427 MPa and 2.0 MPa, respectively. The droplet diameter is set to 4.0 mm, and the droplet velocity is set to 1.0 m/s. The micro-pillar is modeled by the TL-SPH method, and the droplet is modeled by the WC-SPH method. The TL-SPH and WC-SPH models adopt the same initial particle spacing, which is set

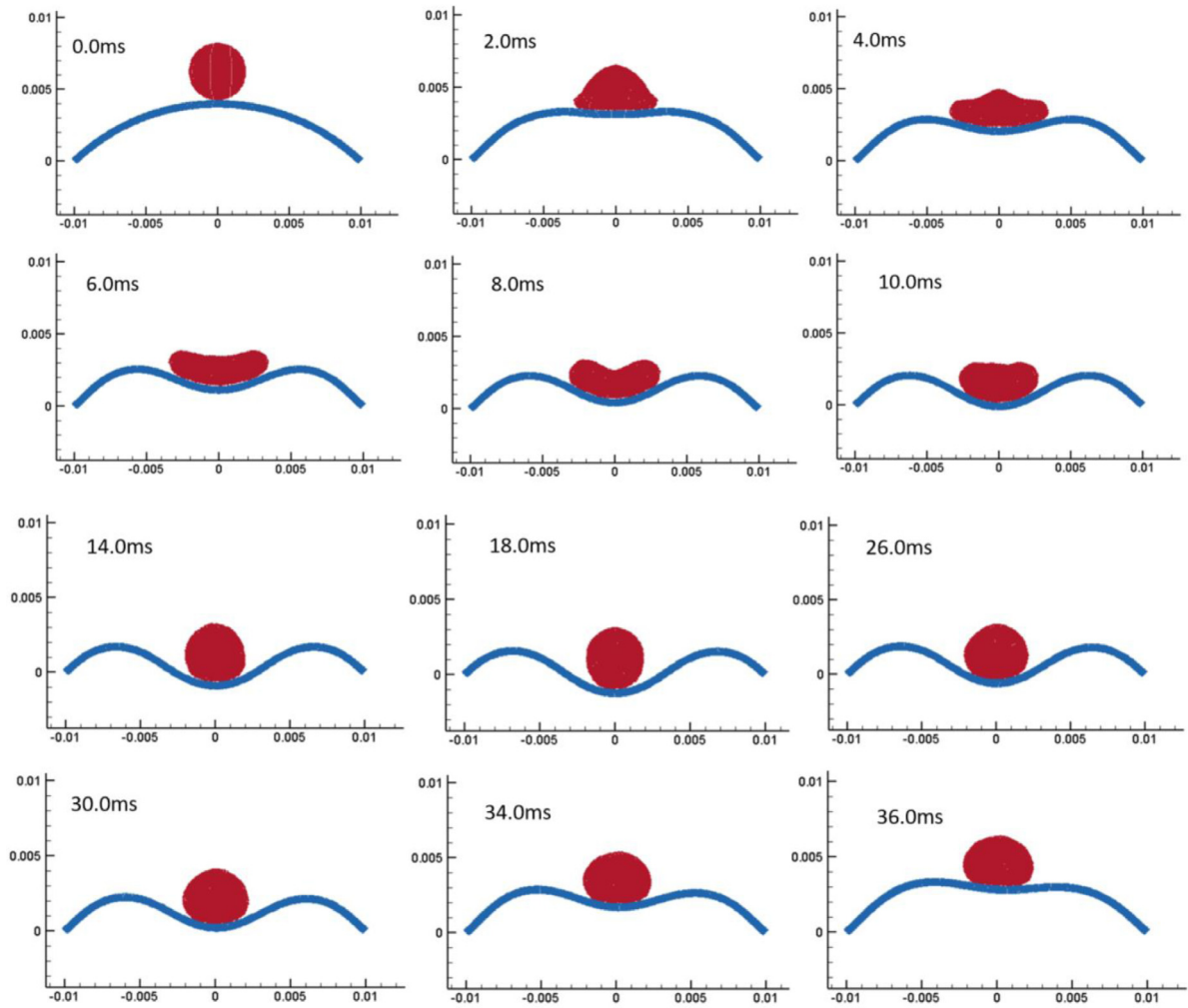


Fig. 18. Simulation results of a droplet impact on a curved beam ($D_i = 4.0$ mm, $V_i = 1.0$ m/s).

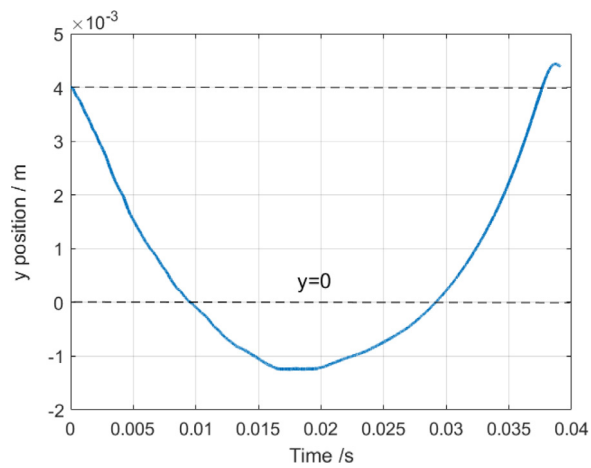


Fig. 19. Time history of the y position of the middle of the beam ($D_i = 4.0$ mm, $V_i = 1.0$ m/s).

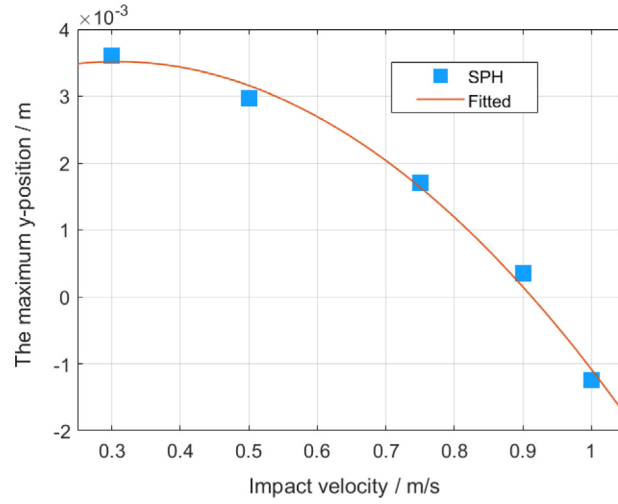
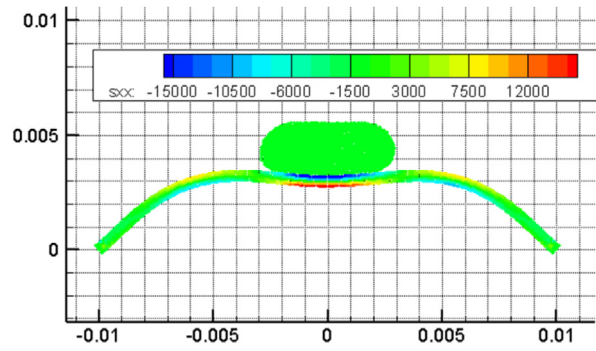
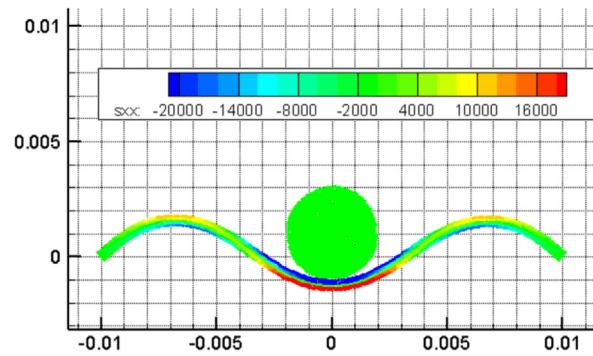


Fig. 20. Relationship between the position at the lowest point of the beam midpoint and the impact velocity.



(a) Velocity 0.5m/s, time 8.0ms



(b) Velocity 1.0m/s, time 19.0ms

Fig. 21. Stress distribution of beam with maximum deformation.

to 0.1 mm. According to the CFL criterion, the maximum time step of WC-SPH should be less than 5.0×10^{-7} s and the maximum time step of TL-SPH model is 1.0×10^{-7} s. Therefore, 1.0×10^{-7} s is chosen as the time step for both WC and TL models.

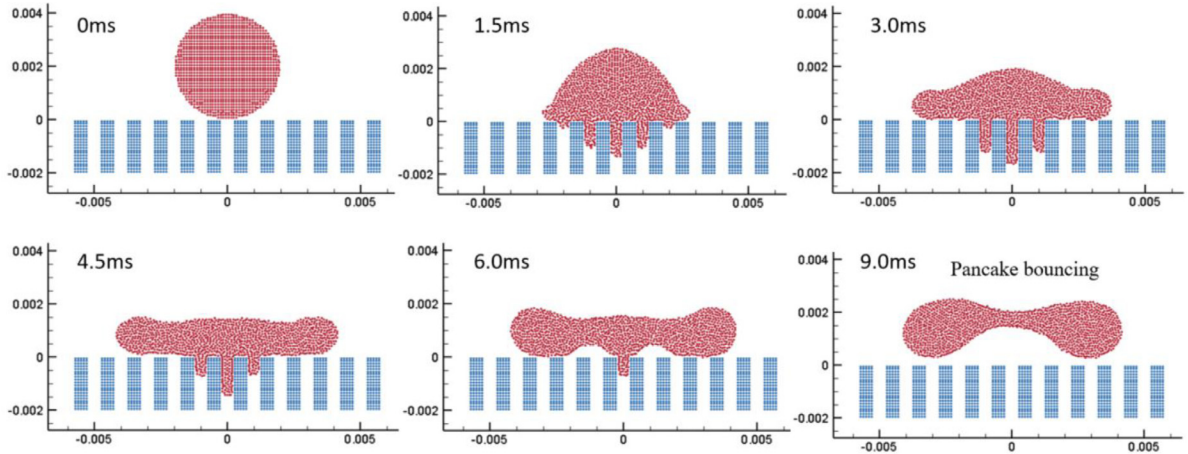


Fig. 22. Simulation results of a droplet impact on a row of rigid pillars ($D_i = 4.0$ mm, $V_i = 1.0$ m/s).

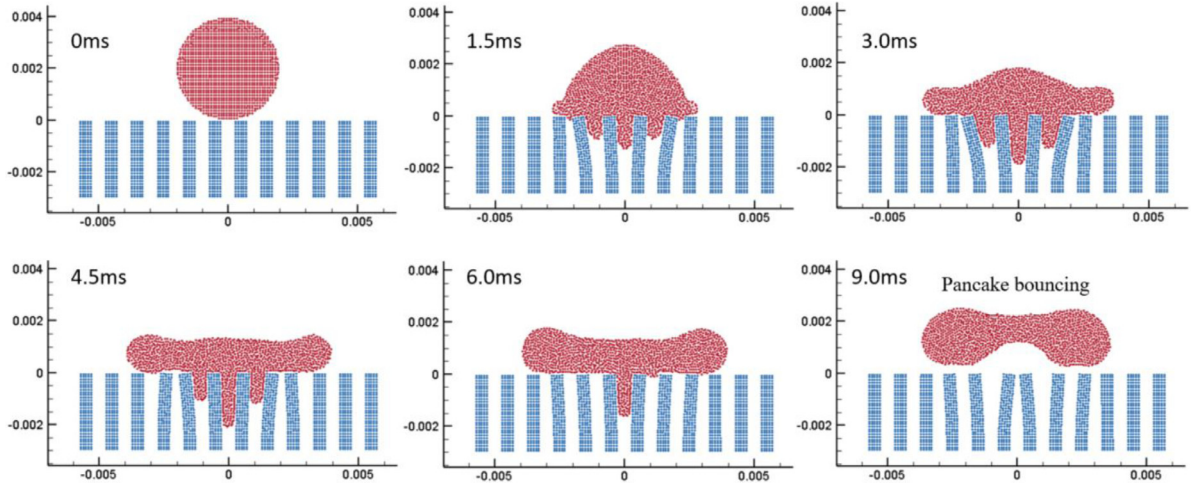


Fig. 23. Simulation results of a droplet impact on a row of flexible pillars ($D_i = 4.0$ mm, $V_i = 1.0$ m/s, $l = 3.0$ mm).

For comparison, the process of droplet impact on a rigid substrate is first simulated. As shown in Fig. 22, the droplet starts to spread after contacting the substrate, and the liquid penetrates into the space between the pillars under the impact pressure (1.5 ms). The liquid penetrates into the three micro-pillar spaces, and the penetration depth at the center of the droplet is the largest. The penetration depth reaches its maximum before the droplet reaches its maximum spreading diameter (3.0 ms). At 4.5 ms, the droplet reaches its maximum spreading diameter while the penetrated liquid has retracted. At 6.0 ms, the droplet at the maximum spreading diameter rebounds as a whole under the retraction of the penetrating liquid, and the process is known as “pancake bouncing”. The liquid that penetrates between the micro-pillars has a greater curvature and therefore higher surface energy, and the liquid retracts faster under the action of surface tension. The infiltrated liquid retracts, causing the droplets omit the retraction process and result in the “pancake bouncing”. The simulation results reveal the mechanism of the pancake bouncing of droplets on the micro-pillar surface, which is the reason to reduce the contact time.

Fig. 23 illustrates the process of droplet impingement on the substrate of a flexible micro-pillar substrate with a length of 3.0 mm, the bottom of which is fixed. Under the action of droplet impact, a single pillar undergoes bending deformation due to the lateral force exerted by the spreading droplet. The bending of the pillar increases the liquid penetration compared to that of a rigid substrate, resulting in an increased retraction time of the penetrated liquid.

As shown in Fig. 23, the deformation of the pillars occurs during the droplet spreading process, with the greatest bending deformation occurring for pillars 3 and 4. This deformation causes the gap between 1 and 3 (2 and 4) to reach its

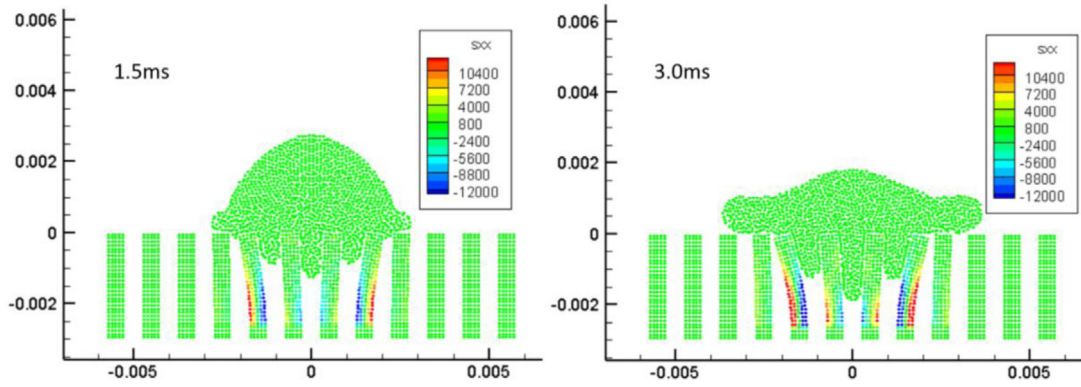


Fig. 24. Stress distribution in the substrate at two time instants ($D_i = 4.0$ mm, $V_i = 1.0$ m/s).

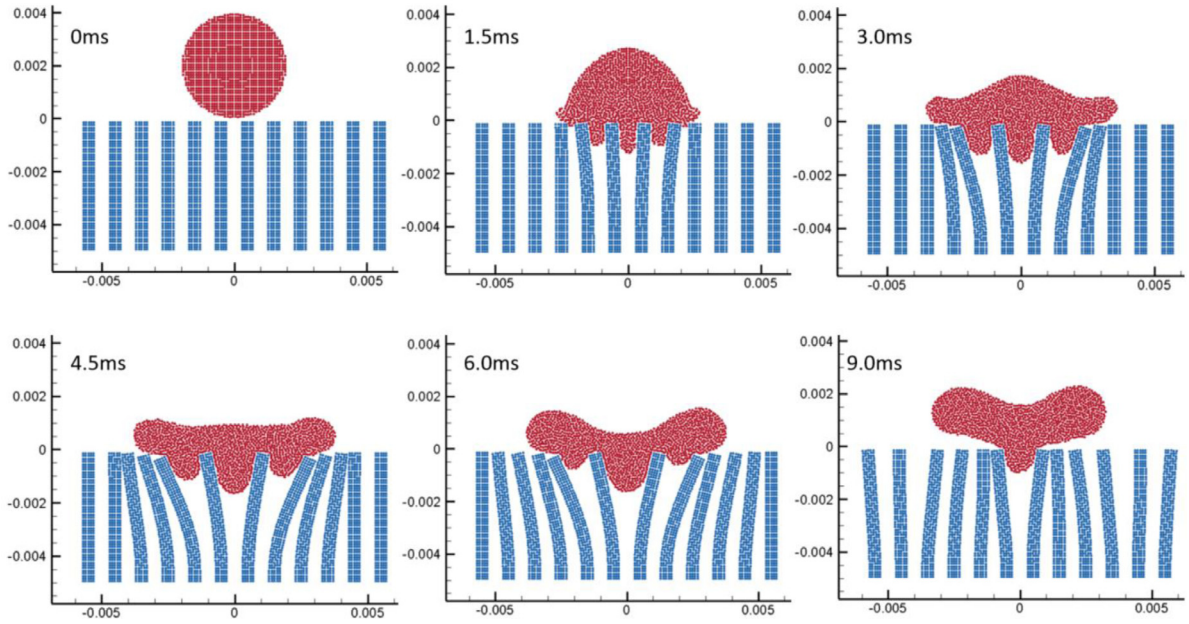


Fig. 25. Simulation results of a droplet impact on a row of flexible pillars ($D_i = 4.0$ mm, $V_i = 1.0$ m/s, $H = 5.0$ mm).

maximum at 3.0 ms, allowing more liquid to penetrate relative to the rigid substrate. The deformed micro-pillar returns to its initial position at about 4.5 ms. Due to the deformation of the micro-pillar, part of the kinetic energy of the droplet is converted into the strain energy, and another part is converted into the surface energy of the droplet. As a result, the maximum spreading diameter of the droplet becomes significantly smaller in comparison to the rigid substrate. The increase of inter-micropillar volume increases the penetrated liquid and also increases the retraction time. At 6.0 ms, the penetration depth is significantly higher than that of the rigid substrate at the same moment. This also leads to a slightly longer contact time than that of the rigid substrate. The droplet still undergoes pancake bouncing on the flexible substrate. The interaction between the micro-pillars is not significant due to their small length (3.0 mm) and the high stiffness.

Fig. 24 shows the stress distribution of the micro-pillars at different moments. Pillars 3 and 4 have higher bending deformation and higher stresses near the bottom end, making them more prone to failure than pillars 1 and 2. Although pillars 5 and 6 also come into contact with the spreading droplets, the resulting deformation is negligible. The simulation results of droplet impact on a longer, flexible micro-pillar substrate are given in Fig. 25. The length of the pillar is set to 5.0 mm. The micro-pillar has less stiffness and produces more bending deformation under the action of a liquid droplet.

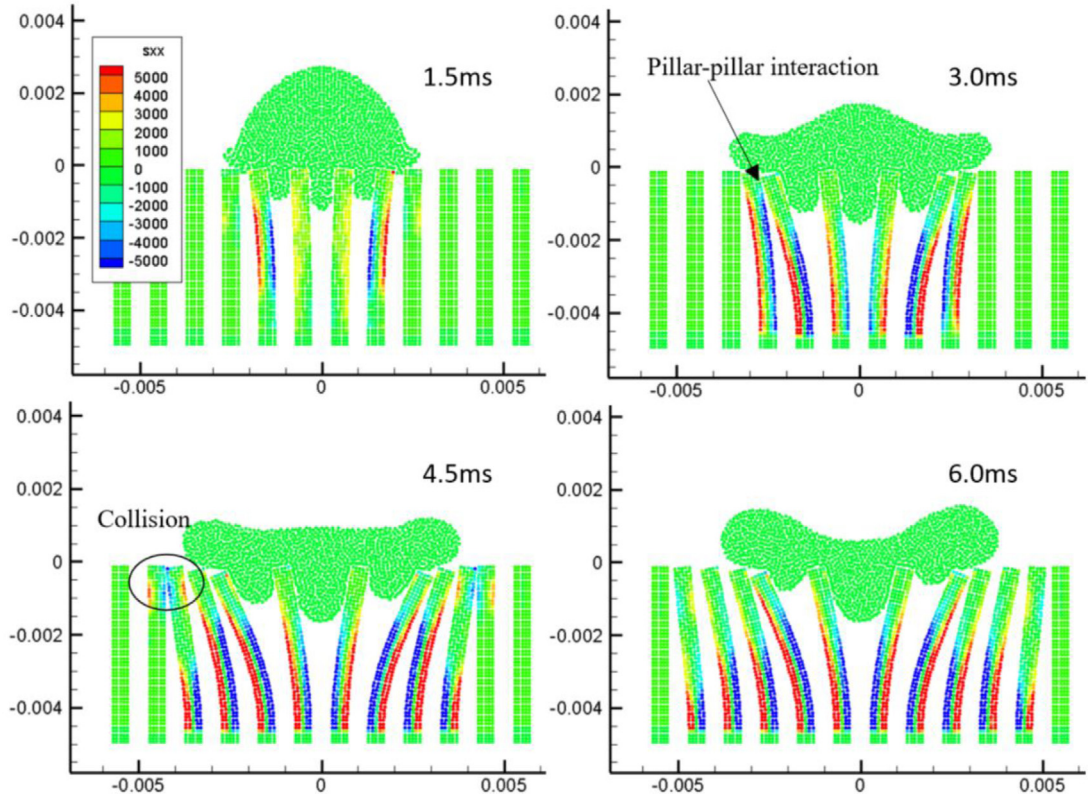


Fig. 26. Stress distribution in the substrate at four time instants ($D_i = 4.0$ mm, $V_i = 1.0$ m/s, $H = 5.0$ mm).

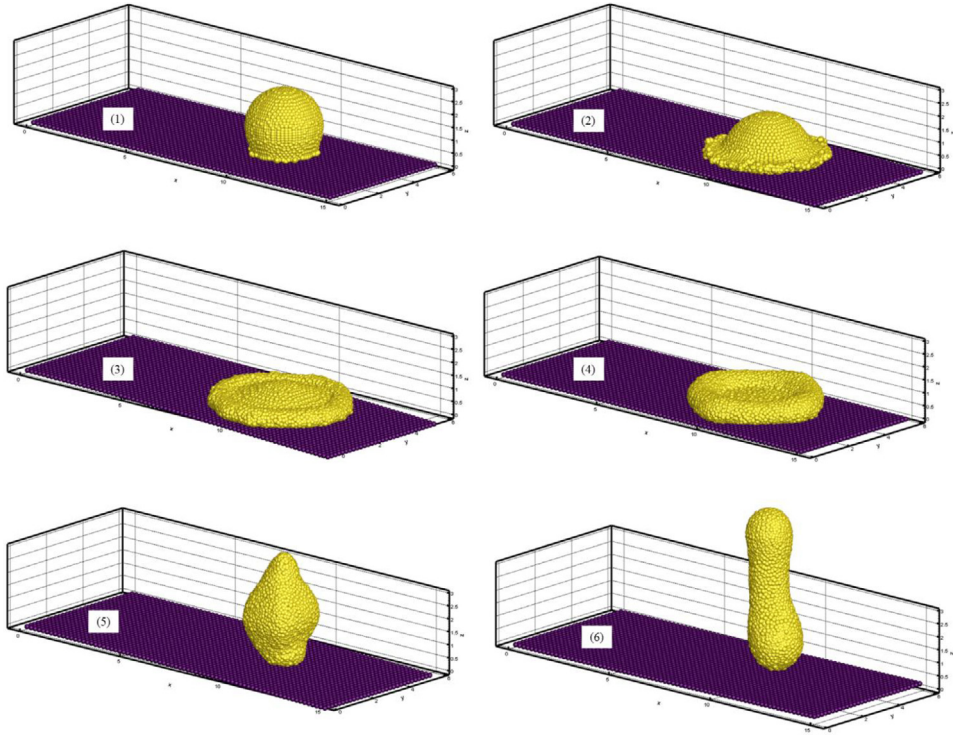
As shown in Fig. 25, the interaction between the micro-pillars is more obvious due to the large deformation of the pillar. At 3.0 ms, pillar 3 contacts pillar 5, causing pillar-5 to bend. At 4.5 ms, pillar 5 contacts pillar 7. The elastic deformation of the micro-pillars absorbs more kinetic energy of the droplet and also further increases the contact time of the droplet. At 9.0 ms, the retraction process of the permeate liquid is not yet complete. As shown in Fig. 26, the region of micro-pillars with bending deformation is significantly larger than that of the 3.0 mm long micro-pillar substrate. The maximum deformation occurs at the maximum spreading diameter of droplet, and the deformation of pillars 3 and 4 is the most severe. The simulation results show that the model can effectively reproduce the interaction between droplet and micro-pillar, and help to explain the dynamic wetting mechanism on the flexible micro-pillar substrate. The model presented in this work can compensate for the previous assumptions of rigid, non-deformable substrates.

5.3. 3D simulation of droplet impact on a cantilever beam

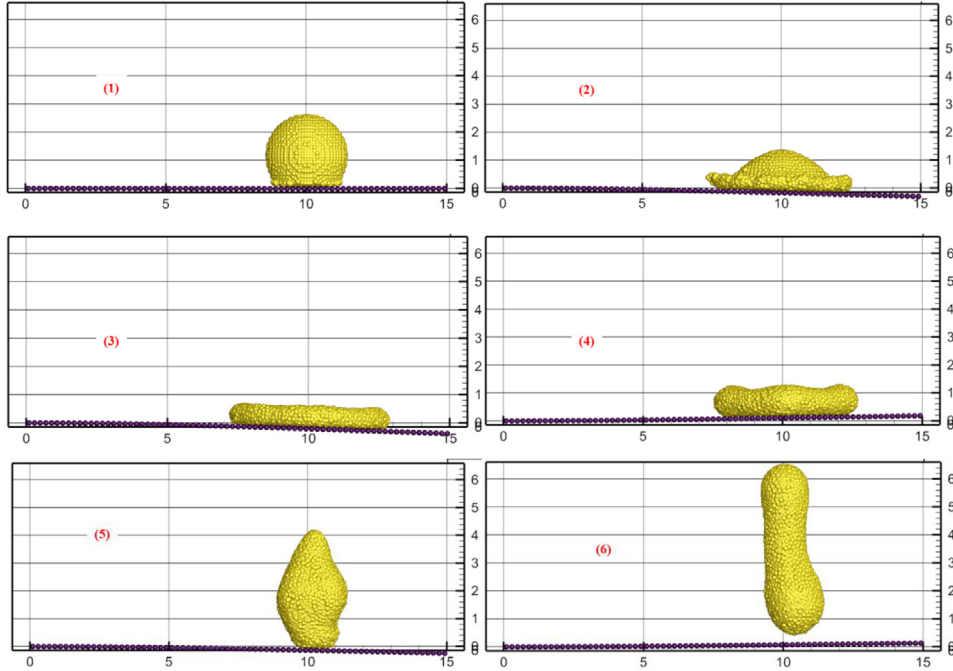
In this section, the 3D SPH model is used to simulate a single droplet impacts on a cantilever beam. The cantilever beam, a slender structure, is modeled by the SPH shell model. The droplet is modeled using WC-SPH model in 3D. The WC-SPH model established in Section 2 is applicable to both 2D and 3D, with corresponding kernel function coefficients selected.

Droplet impact can cause elastic deformation of the cantilever beam, during which part of the kinetic energy of the droplet is converted into the strain energy of the beam. The impact process contains two energy conversion mechanisms, i.e., kinetic-to-surface and kinetic-to-elastic energy (Weissensee et al., 2017). The two mechanisms occur during the contact between the droplet and the surface of the cantilever. Once the droplet bounces off the surface, the cantilever will enter a free vibration state. If the cantilever reaches the maximum deformation before the droplet rebounds, the deformed cantilever will give the droplet a upward force during the recovery process, which makes the droplet bounce off the surface earlier. This is the so-called springboard effect through which an elastic substrate reduces the droplet contact time $\square$.

To simulate this springboard effect, the simulation parameters are configured as following. The dimensions $l \times b \times t$ of the cantilever is set to $15 \times 6 \times 0.3$ mm, and the corresponding stiffness of the beam is $\mathbf{K}_b = 6.0$ N/m. The droplet



(a) Isometric view



(b) Front view

Fig. 27. Snapshots of simulation and experiment of droplet impact on cantilever ($l = 15$ mm, $K_b = 6.0$ N/m).

diameter is set to 2.8 mm and the impact velocity is set to 1.4 m/s, corresponding to $We = 76.2$. The impact point is 5.0 mm from the end of the cantilever. Fig. 27 shows the snapshots at different time instants. As the droplet contacts the

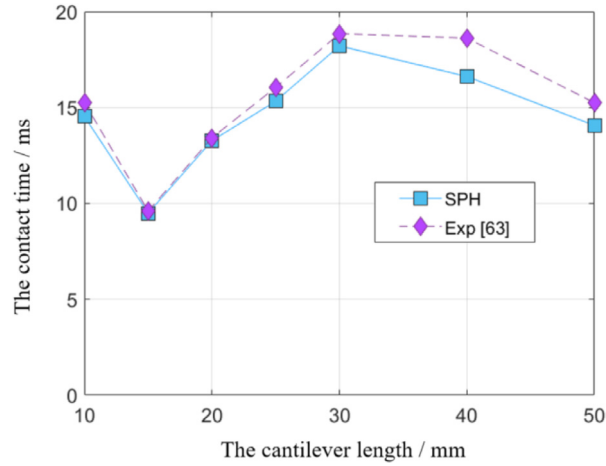


Fig. 28. The relationship between contact time and cantilever length was compared between simulation and experimental results ($D_i = 2.8$ mm, $V_i = 1.4$ m/s).

cantilever, it spreads on the surface, and the impact force causes the cantilever to deform (Fig. 27(a)(b)). As the droplet reaches the maximum spreading diameter, the end of the cantilever also moves to the maximum downward position (Fig. 27(c)). Then, the droplet starts to retract, and the cantilever beam starts to recover. The cantilever gives the droplet an upward force, which makes the droplet detaches the surface earlier. The droplet bounces off the surface of the cantilever in about 9.0 ms (Fig. 27(d)). The contact time is reduced by 40% compared to the rigid substrate under the same impact conditions. Simulation results of droplet rebound behavior and the cantilever end displacement are in good agreement with the experimental results which are also given in the figure.

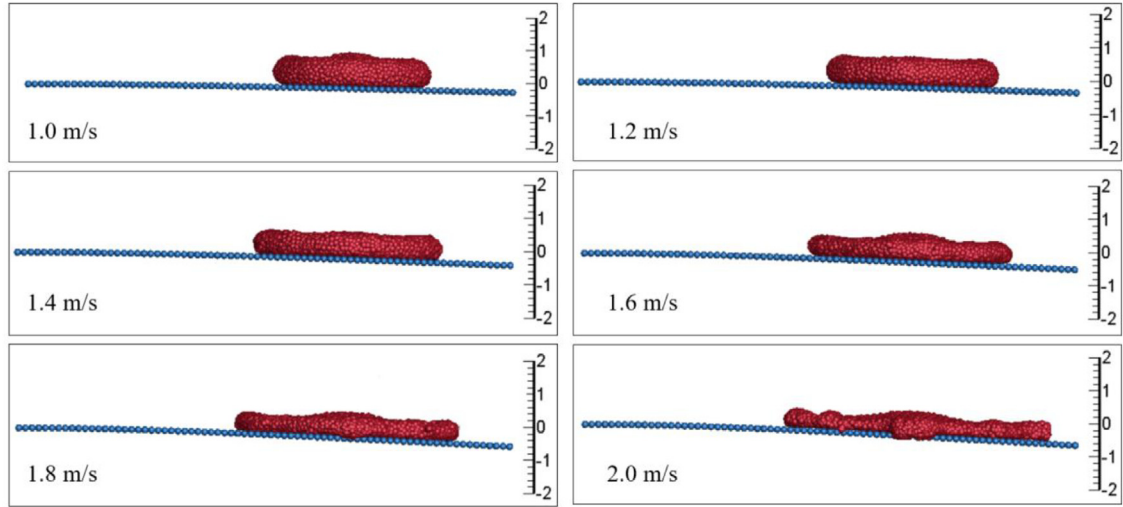
Fig. 28 shows the effect of beam length on droplet contact time. When the beam length is 15 mm, under the action of the springboard effect, the contact time of the droplet reaches the minimum, which is about 60% of the contact time of the rigid substrate. When the beam length is 10 mm, the stiffness of the beam is large enough to be like a rigid substrate, the contact time is very close to the rigid substrate. When the beam length is 30 mm, the contact time of the droplet reaches a maximum value, which is 15% higher than that of the rigid substrate. Fig. 30 shows the simulation results of the droplet impacting the cantilever with the length of 30 mm. When the droplet reaches the maximum spreading diameter (3.4 ms), the cantilever is still moving downward. When the end of the cantilever reaches the maximum displacement, the droplet has shrunk and is ready to rebound (11.4 ms). Subsequently, the cantilever beam also begins to recover, and the end of the beam moves upward. The upward motion of the beam makes the beam surface always in contact with the droplet, delaying the detachment of the droplet, ultimately increasing the contact time.

The deformation of the cantilever is related to the magnitude of the droplet impact force. The impact force depends on the inertial parameters of the droplet, including the droplet diameter D_i and impact velocity V_i . The impact force of the droplet on the solid surface during the spreading and retracting process is not constant, it has a trend of increasing first and then decreasing (Soto et al., 2014), that is, there exists a peak value F_0 . The droplet impact force F_0 is expressed as $F_0 = \pi \rho R^2 V^2$. It indicates that the impact force is proportional to the square of the velocity. Soto et al. (2014) derived the theoretical equation for the maximum displacement of the end of the cantilever beam as:

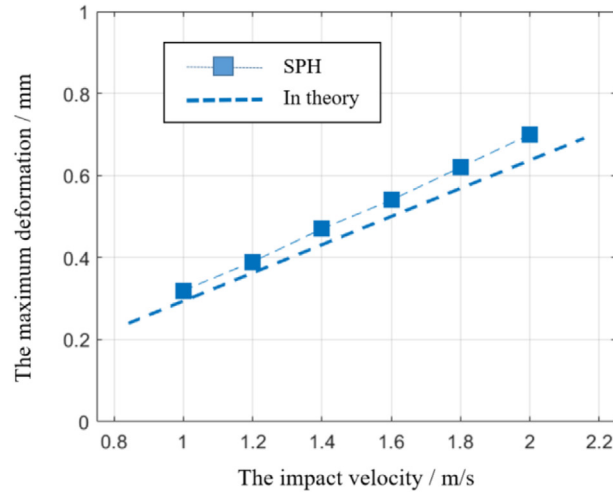
$$\delta_{\max} = \frac{3}{2\pi} \cdot \frac{m}{M} \cdot \frac{V_i}{f_b} \quad (95)$$

where M is the mass of the cantilever, f_b is the vibration frequency of the cantilever, m is the mass of the droplet. Eq. (95) is suitable for the case where the characteristic time of the cantilever is greater than the contact time of the droplet, where the maximum displacement is linear with the impact velocity.

The proposed SPH model is used to simulate the process of a 2.8 mm droplet impact on a cantilever beam with the length 15 mm at different impact velocities. As shown in Fig. 29(a), when the cantilever beam reaches the maximum deformation, the droplet just reaches the maximum spreading diameter, which indicates that the characteristic time of the droplet and the cantilever beam are matched. Simulation results show that this match is not affected by impact velocity. As shown in Fig. 29(b), the maximum displacement of the cantilever obtained by simulation has a linear relationship with the impact velocity, which is consistent with analytical Eq. (95).



(a) Simulation results using different impact velocities. Each snapshot corresponds to the momentum when the cantilever at the maximum deformation position.

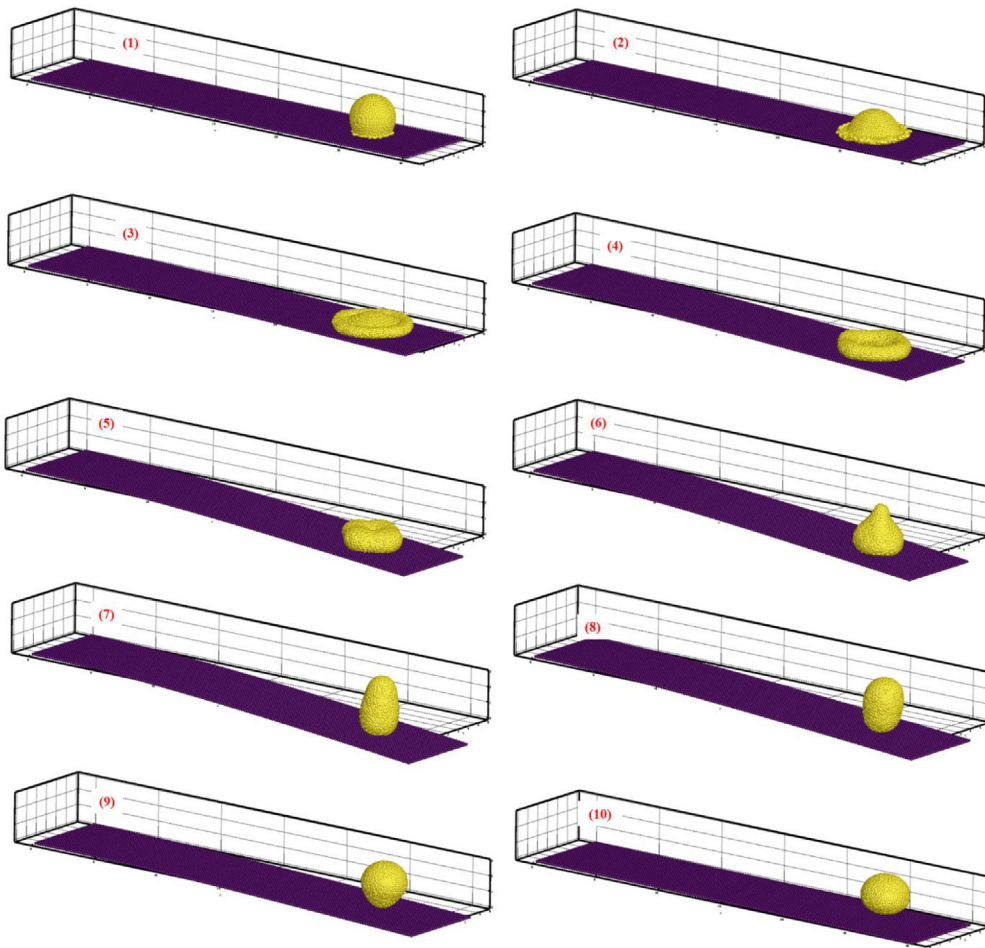


(b) Comparison of maximum deflection of the cantilever between SPH and analytical results.

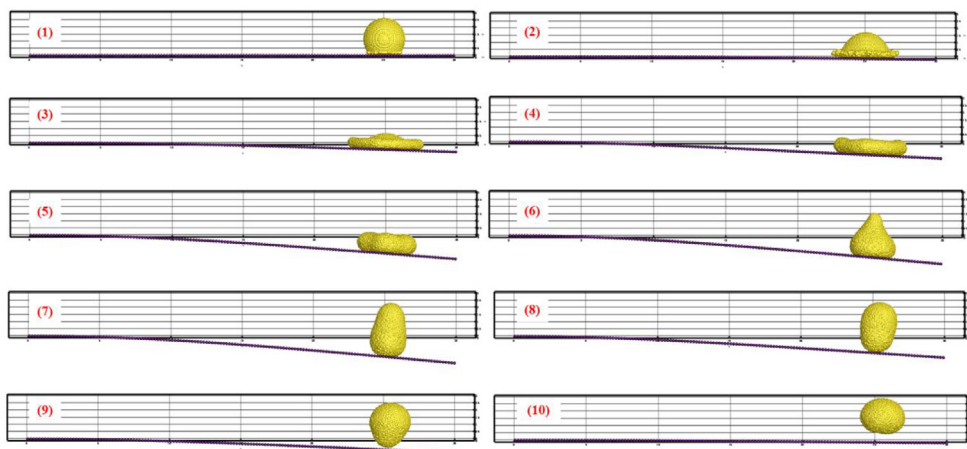
Fig. 29. Effect of impact velocity on the deformation of the cantilever ($l = 15.0$ mm).

6. Conclusion

The meshfree smoothed particle hydrodynamics (SPH) method is used to establish a fluid–solid coupling model for the simulation of liquid droplets impacting flexible structures. The model is used to simulate the dynamic process of droplets impacting several types of flexible substrates. In the model, both the fluid (*i.e.*, the droplet) and the solid (*i.e.*, the substrate) are described by SPH. The weakly compressible SPH method is used to establish the droplet impact model, and the total Lagrangian SPH method is used to establish the solid model. This study presents corresponding modeling schemes for both two-dimensional and three-dimensional problems. The surface tension of the droplet is simulated by the CSF method, and the calculation of the surface normal vector and curvature is realized by surface reconstruction. This study also establishes algorithms for surface particle detection and interpolation reconstruction for both 2D and 3D problems. For the simulation of flexible substrates, the TL-SPH model is directly used to establish the solid model in 2D cases. In 3D cases, in order to improve the computational efficiency of simulating slender flexible substrates, this study extends the TL-SPH model and establishes the TL-SPH shell model.



(a) Isometric view



(b) Front view

Fig. 30. Snapshots of simulation and experiment of droplet impact on cantilever ($l = 30$ mm, $K_b = 0.8$ N/m).

CRediT authorship contribution statement

Xiangwei Dong: Conceptualization, Methodology, Software, Writing – review & editing, Funding acquisition. **Yanxin Liu:** Data curation, Writing – original draft. **Ran Yu:** Visualization, Investigation. **Menghao Fan:** Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfluidstructs.2023.103864>.

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